

Interleukin 10 controls the balance between tolerance, pathogen elimination and immunopathology in birds

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IL-10 balances protective and deleterious immune functions in mice and humans, but if IL-10 also controls avian intestinal homeostasis remains unclear. Generating genetic knockouts, Meunier et al. established that a complete lack of IL-10 strengthened immunity against enteric bacteria in chickens, while also aggravating infection-inflicted inflammatory tissue damage and dysbiosis upon parasite infection, but unlike mouse models, IL-10 deficiency did not provoke spontaneous colitis in chickens. The findings presented are **valuable**, and the strength of evidence is **convincing**. The observation may have implications for the livestock industry and additional studies involving genetic knockouts may further unravel conserved and distinct avian IL-10 controls.

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Abstract

Effective mucosal immunity in the intestine involves a fine balance between tolerance of the microbiome, recognition and elimination of pathogens, and inflammatory tissue injury. The anti-inflammatory cytokine IL10 regulates these processes in the intestines of mice and humans; the anti-inflammatory activity of IL10 is also conserved in birds. To determine the function of IL10 in avian mucosal immunity, we generated germ line modifications of the chicken *IL10* locus to abolish or reduce IL10 expression. *In vitro* analysis of macrophage response to lipopolysaccharide confirmed the loss of IL10 protein expression, the lack of dosage compensation in heterozygotes, and prevention of autocrine inhibition of nitric oxide production in homozygous IL10 knockout macrophages. IL10-deficiency significantly altered the composition of the caecal microbiome, but unlike IL10-deficient mice and humans, IL10-deficient chickens did not exhibit spontaneous colitis. Following experimental challenge with

Salmonella enterica serovar Typhimurium or *Campylobacter jejuni* in IL10-deficient chickens, enhanced clearance of the pathogens was associated with elevated transcription of pro-inflammatory genes and increased infiltration of inflammatory cells into gut mucosa. In IL10-deficient chickens challenged with the parasite *Eimeria tenella*, pathogen clearance was accelerated but caecal lesions were more severe and weight gain was compromised. Neither the heterozygous IL10 knockout nor a homozygous IL10 enhancer mutation had a major effect on pathogen clearance or inflammation in any of the challenge models. Our findings highlight the intrinsic compromise in mucosal immune response and have important implications for the development of strategies to combat avian and zoonotic pathogens in poultry.

Introduction

Elimination of potential pathogens by the innate and acquired immune systems involves an intrinsic compromise between host protection and tissue injury that requires mechanisms to control the nature, magnitude, duration and specificity of effector pathways. A key component of specificity is the capacity to distinguish between pathogens and the commensal microbiome, and nowhere is this challenge more pressing than in the intestine. The anti-inflammatory cytokine, interleukin 10 (IL10), has long been recognized as an essential feedback regulator of intestinal inflammation in mammals (reviewed in [1]). IL10-deficient mice generated by homologous recombination developed spontaneous enterocolitis [2]. Subsequent studies revealed that components of the intestinal microbiome were essential to development of pathology and determined the extent and location of lesions [3]. In humans, homozygous mutations in genes encoding IL10 or its receptor, IL10RA, have been associated with early-onset inflammatory bowel disease [1].

In mammals, IL10 is produced by both myeloid and T and B lymphoid cells following activation by immune stimuli [1, 4], with subtle differences between mice and humans [5]. In the specific context of the intestine, conditional deletion of the *IL10* and *IL10RA* genes in mice indicated that production of the ligand by macrophages was not required for pathology, consistent with T cells being the major source of intestinal IL10 [6, 7]. By contrast, macrophage-specific deletion of the *IL10RA* gene encoding the IL10 receptor was sufficient to elicit spontaneous colitis in mice [8–10]. The contribution of IL10 to feedback modulation of host defense is not restricted to the gut. There have been countless studies of the impact of IL10 deficiency in mouse parasite, bacterial, fungal and viral disease models (reviewed in [4]). The primary focus of these studies has been on the mitigation of immunopathology by IL10, with less attention given to pathogen elimination. By contrast to studies in mice, analysis of the quantitative importance of IL10 in other species has been largely correlative. Both ligand and receptor are conserved in vertebrates, and zebrafish has been discussed as a model for understanding IL10 biology [11]. Here we focus on the chicken, both a model vertebrate and an economically important livestock species.

The original cloning and sequencing of chicken *IL10* cDNA and genomic DNA [12] described the immunomodulatory activity of the recombinant protein on T cell production of interferon-gamma (IFN γ), its regulated expression in T cells and macrophages *in vitro*, and increased mRNA expression during parasite (*Eimeria maxima*) infection. Like the mammalian transcript, chicken *IL10* mRNA contains multiple copies of the AUUUA instability motif in the 3'-untranslated region (UTR). We and others have provided evidence that quantitative variation in IL10 production may contribute to inter-individual or inter-breed variation in infectious disease susceptibility in chickens [13–17]. To test this hypothesis directly, and more generally to determine the conservation of IL10 biological function across vertebrates, we have generated germ line modifications of the chicken *IL10* locus that abolish or reduce IL10 expression. We describe the

lack of dosage compensation in heterozygous mutation, and present evidence that the reduced expression of IL10 leads to increased pathogen clearance at the expense of increased inflammation.

Results

Generation of IL10-deficient chickens

The emergence of CRISPR/Cas9 technology and the ability to culture, cryopreserve, genetically modify and subsequently transfer chicken primordial germ cells (PGCs) into sterile surrogate host embryos has expedited the analysis of gene function in the chick (**Figure 1A**; [18–21]). We used this approach to edit the *IL10* locus in chicken PGCs and create two independent lines of IL10-deficient chickens (**Figure 1**). We first created an IL10 knockout (IL10KO) line by introducing a premature in-frame stop codon in exon 1 of the *IL10* gene in chicken PGCs, 75 bp downstream of the start codon (C26>*; **Figure 1B** and **D**, and SI Appendix, Figure S1). To generate a potential expression hypomorph, we created an IL10-Enhancer knockout (IL10EnKO) line by deleting a 533-bp genomic fragment encompassing a non-coding sequence located approximately 2.3–2.8 kb upstream of the *IL10* transcription start site, in chicken PGCs (**Figure 1B–C** and SI Appendix, Figure S2). This sequence was targeted based upon alignment of sequences from multiple avian species that identified a core of 300 bp with conservation around 85%, containing perfectly conserved motifs for known transcriptional regulators, including AP1, FOXP3 and members of the NFAT/ETS family (SI Appendix, Figure S3). The size and species conservation of this element resembles the well-characterized intronic enhancer of the *Csf1r* locus [22].

To create IL10KO and IL10EnKO chickens, male PGCs carrying the desired biallelic edit in *IL10* exon 1 or in the putative *IL10* enhancer region (as validated by PCR and DNA sequencing; **Figure 1C–D** and SI Appendix, Figure S4) were injected into iCaspase9 sterile surrogate host embryos, in which a chemically inducible caspase 9 protein is expressed from the PGC-specific *DAZZL* locus, together with a GFP reporter [23]. Embryos were treated with the iCaspase9 activation chemical AP20187, incubated to hatch, and the resulting chicks were raised to sexual maturity under specified pathogen-free (SPF) conditions at the National Avian Research Facility (NARF). Mating of the iCaspase9 surrogate host cockerels to Rhode Island Red (RIR) hens generated 100% IL10KO or IL10EnKO heterozygous (HET) offspring in the first generation (G1; **Figure 1A**; see SI Appendix, Table S1 for detailed G1 bird numbers). Upon sexual maturity, G1 HET birds were intercrossed to generate the second generation (G2) of IL10KO and IL10EnKO wildtype (WT), HET and homozygous (HOM) birds for analyses. Both lines produced WT, HET and HOM offspring at a 1:2:1 mendelian ratio (see SI Appendix, Table S1 for detailed G2 bird numbers).

Confirmation of the IL10 mutation in bone marrow-derived macrophages

Chicken bone marrow-derived macrophages (BMDMs) cultured in recombinant macrophage colony-stimulating factor respond to bacterial lipopolysaccharide (LPS) by secreting IL10 that can be detected by ELISA [24]. The autocrine function of IL10 in this assay was demonstrated by the addition of neutralizing anti-IL10 antibody, which amplified the expression of nitric oxide synthase (NOS2) and production of nitric oxide [24]. **Figure 2A** confirms the release of IL10 by LPS-stimulated BMDMs derived from WT embryos and the complete absence of detectable IL10 release by BMDMs derived from IL10KO HOM embryos. The absence of IL10 was associated with a significant increase in nitric oxide production that was not additive with the impact of added anti-IL10 antibody (**Figure 2B**). Since IL10 mediates feedback control it is conceivable that the loss of one functional allele could be compensated by increased expression from the other allele. However, we found that expression of IL10 was reduced by up to 50% in BMDMs derived from

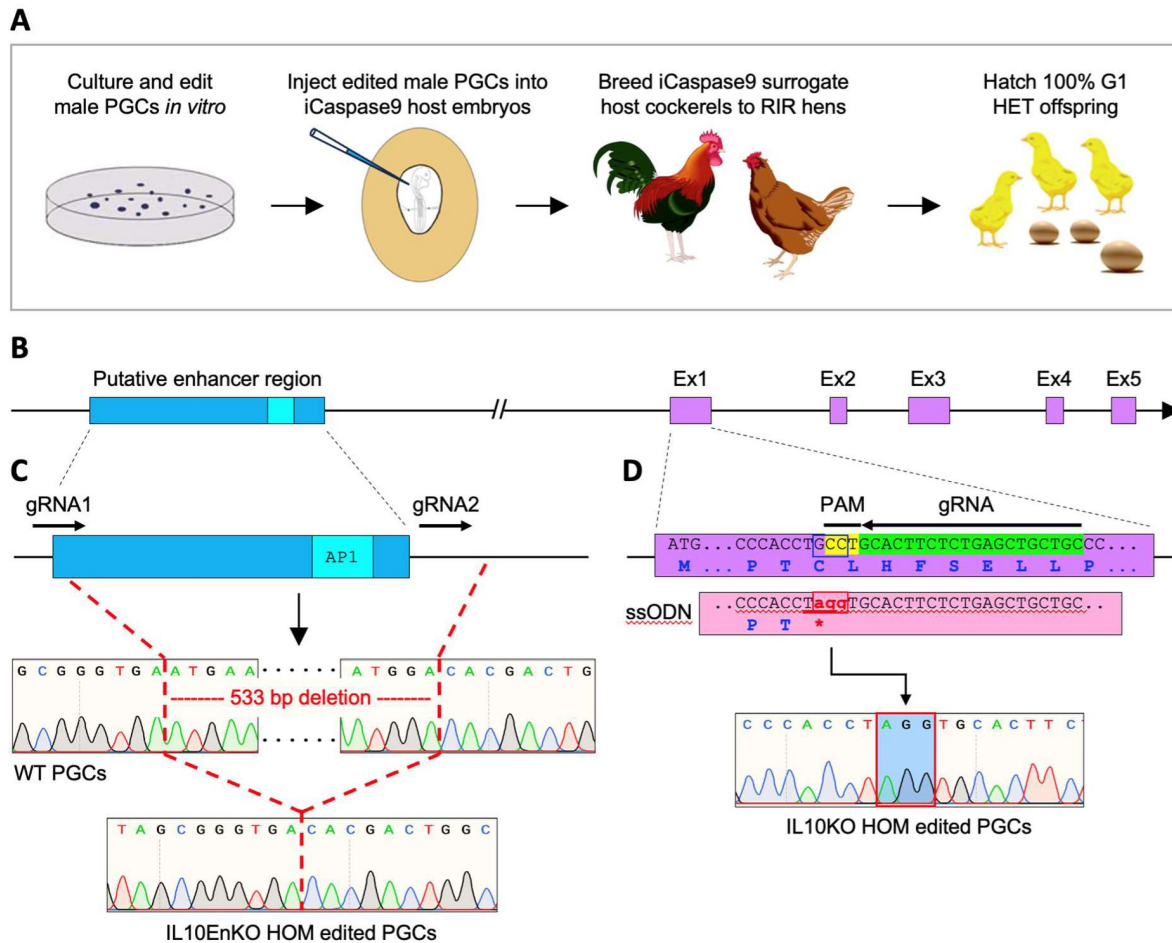


Figure 1

IL10KO and IL10EnKO editing strategies.

A: Schematic of the pipeline used to create gene-edited chickens using embryonic PGCs and the iCaspase9 sterile surrogate host line. Male PGCs carrying the desired biallelic (HOM) edits were injected into iCaspase9 sterile surrogate host embryos to create IL10KO and IL10EnKO chickens. Upon sexual maturity, surrogate host cockerels were bred to RIR hens, resulting in 100% IL10KO HET or IL10EnKO HET offspring in the first generation (G1). **B:** Schematic of the *IL10* locus depicting the *IL10* putative enhancer region and the five *IL10* exons (Ex1 to Ex5). **C:** IL10EnKO HOM PGCs were created by deleting a 533-bp fragment encompassing the *IL10* putative enhancer region, using two gRNAs; DNA sequencing confirmed the biallelic deletion. **D:** IL10KO HOM PGCs were created using one gRNA and a 143-bp repair template (ssODN) to modify three nucleotides in *IL10* exon 1 (red, lowercase), thus introducing a premature in-frame stop codon in the *IL10* gene (underlined, *); DNA sequencing confirmed the biallelic edit.

IL10KO HET embryos, with a corresponding partial impact on nitric oxide production (**Figure 2A-B**). The impact of the IL10EnKO HOM mutation was more variable between samples in both assays, but consistent with a partial expression hypomorph (**Figure 2A-B**).

Loss of IL10 does not lead to spontaneous immunopathology

The IL10KO and IL10EnKO chicken lines were created and raised under SPF conditions in the NARF SPF avian facility. Close monitoring of the WT, HET and HOM birds over several months in that environment did not reveal any health issues or adverse phenotypes. The IL10KO and IL10EnKO edits did not significantly affect the post-hatch growth of HET and HOM birds compared to their respective WT controls (**Figure 3A-B**), although IL10EnKO HET and HOM hens were overall heavier than IL10EnKO WT controls (**Figure 3B**). Regular gross post-mortem examination of tissues and histopathological analyses of the gastrointestinal tract over several months (using a semiquantitative scoring system [25]) revealed no evidence of intestinal pathology in IL10KO HOM and IL10EnKO HOM birds compared to WT controls (**Figure 3C**). These results are in contrast with the early onset colitis and failure to thrive phenotypes observed in IL10-deficient mouse models in both conventional and SPF environments [2, 9, 26, 27].

Spontaneous colitis in *Il10*^{-/-} mice is associated with and dependent upon dysbiosis in the intestinal microbiome [28–30]. We compared the caecal microbiome of IL10KO WT and HOM birds at 4 weeks of age using 16S rDNA V3-V4 hypervariable amplicon sequencing as described [31, 32]. The caecal microbiome of both genotypes was dominated by the phylum *Firmicutes* (SI Appendix, Figure S5A) and the most abundant genera were *Lactobacillus* and *Faecalibacterium* (SI Appendix, Figure S5B). Linear discriminant analysis (LDA) Effect Size (LEfSe) analysis showed enrichment of the phylum *Firmicutes* and the genera *Deffluviitaleaceae_UCG-011* and *Paludicola* in the IL10KO HOM group, while the phylum *Bacteroidota* and the genera *Alistipes* and *UCG_008* were enriched in the WT group (SI Appendix, Figure S5C). Principal component analyses showed clear separation of the microbial communities based on beta diversity (SI Appendix, Figure S5D, $R^2 = 0.25$, $P = 0.017$). These findings confirm that mucosal IL10 regulates the composition of the chicken microbiome.

To evaluate a potential environmental impact on the phenotype of IL10-deficient birds, a cohort of IL10KO WT and HOM birds was hatched and raised in the NARF conventional avian facility. IL10KO HOM birds remained healthy in that environment as they aged. Post-growth hatch rates were comparable between IL10KO WT and HOM hens, with the exception of a marginal effect on growth that resolved with age (**Figure 3D**); there was also no evidence of overt intestinal pathology in IL10KO HOM birds compared to WT controls (**Figure 3E**). Cellular infiltration scores were however significantly higher in the duodenum and proximal colon of both IL10KO WT and HOM birds raised in the conventional facility as opposed to the SPF environment (**Figure 3F**).

Two important differences exist between the IL10KO mouse and chicken models. Firstly, the birds are outbred. Most mouse studies are conducted on an inbred background and the impact of the *IL10* mutation is strain-specific [27]. Secondly, chickens in the NARF conventional avian facility are routinely immunised against major avian pathogens (SI Appendix, Table S2). The original analysis of IL10-deficient mice revealed little impact on T cell-dependent antibody responses [2]. We measured antibody titres in 29-weeks old IL10KO WT and HOM birds raised in the NARF conventional facility for 7 of the pathogens that the birds were vaccinated against (SI Appendix, Figure S6): avian encephalomyelitis virus (AEV), chicken anaemia virus (CAV), duck adenovirus (the agent of egg drop syndrome, EDS), infectious bursal disease virus (IBDV), infectious bronchitis virus (IBV), infectious laryngotracheitis virus (ILTV), and Newcastle disease virus (NDV). The administration schedule, nature and delivery mode of these vaccines is summarised in SI Appendix, Table S2.

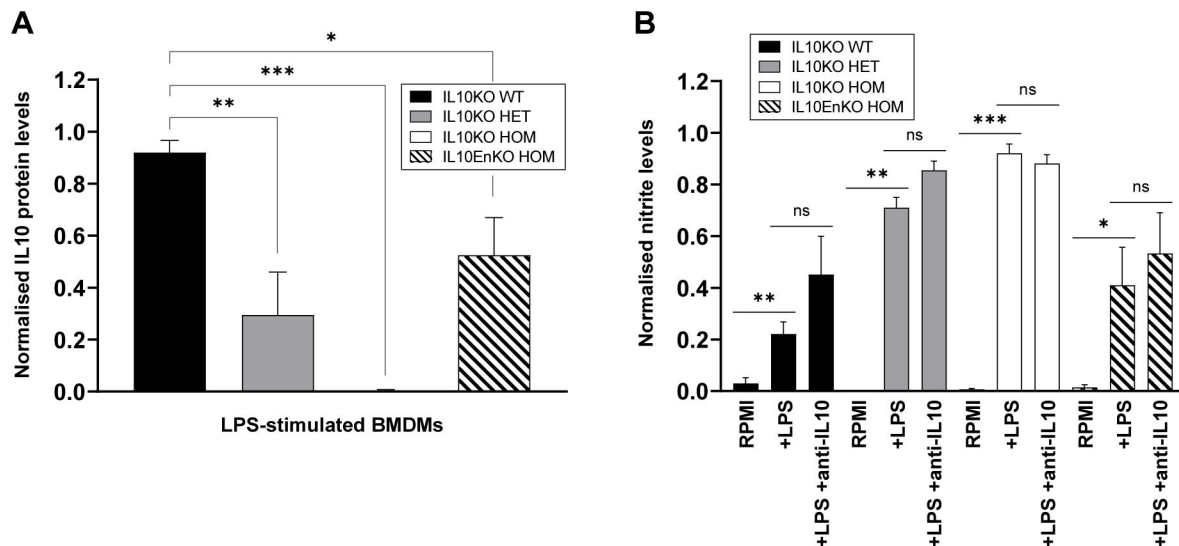


Figure 2

Confirmation of the IL10 mutation in bone marrow-derived macrophages.

A: IL10 protein levels were measured by capture ELISA for LPS-stimulated BMDMs derived from Day 18 IL10KO WT (n=4), HET (n=2), HOM (n=4) and IL10EnKO HOM (n=4) embryos. Two independent experiments were performed and protein levels were normalised against the highest IL10KO WT value for each experiment and then combined. LPS treatment induced IL10 production in IL10KO WT BMDMs, but not in IL10KO HOM BMDMs; intermediate levels of IL10 were detected in IL10KO HET and IL10EnKO HOM BMDMs; IL10 expression in non-LPS induced samples was negligible (<0.015 ; not shown). **B:** Nitric oxide production was assessed by measuring nitrite levels using Griess assay for LPS-stimulated BMDMs derived from Day 18 IL10KO WT (n=6), HET (n=2), HOM (n=7) and IL10EnKO HOM (n=6) embryos, in the absence or presence of neutralizing anti-IL10 antibody ROS-AV163 (anti-IL10). Three independent experiments were performed and nitrite levels were normalised against the highest IL10KO HOM nitrite value for each experiment before being combined. A significant increase in nitrite levels was observed in the absence of IL10 in LPS-stimulated IL10KO HOM BMDMs; nitrite levels were also significantly increased in LPS-stimulated IL10KO HET and IL10EnKO HOM BMDMs. Addition of neutralizing anti-IL10 antibody did not significantly affect nitrite levels in LPS-stimulated BMDMs, independently of the IL10 genotype. Note that each BMDM sample was derived from three pooled embryos of the same genotype. Data displayed as mean with SE. Statistical significance calculated using two-tailed unpaired t tests; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, ns: not significant.

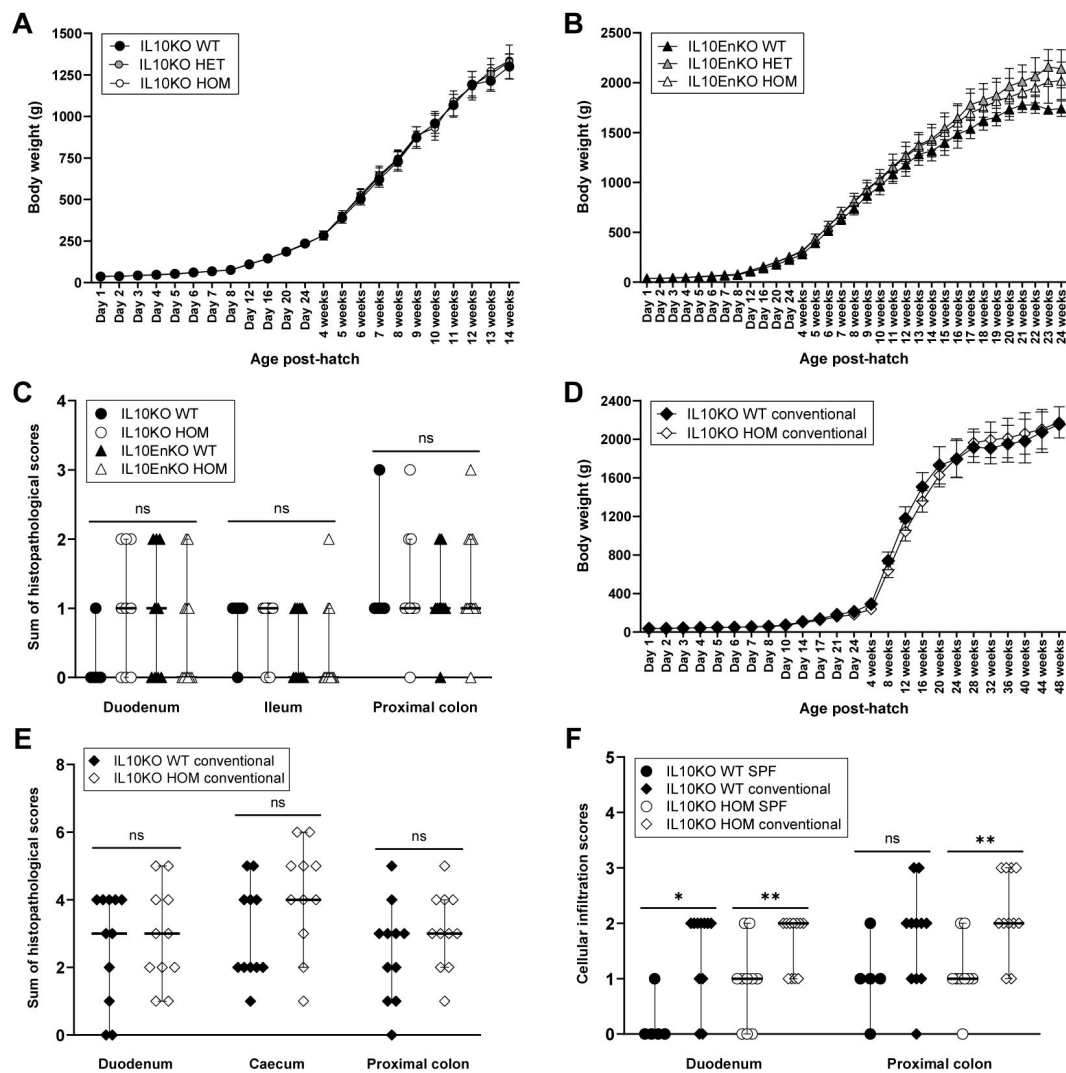


Figure 3

Growth curves and histopathological analyses of the gastrointestinal tract of IL10-deficient chickens in SPF and conventional facilities.

A: Growth curves for IL10KO WT ($n=3$ to 8), HET ($n=11$ to 16) and HOM ($n=2$ to 4) hens raised in the NARF SPF facility, from hatch to 14 weeks. No significant weight differences were observed between genotypes at any timepoint ($P > 0.05$). **B:** Growth curves for IL10EnKO WT ($n=2$ to 4), HET ($n=7$ to 11) and HOM ($n=5$ to 11) hens raised in the NARF SPF facility, from hatch to 24 weeks. IL10EnKO HET and HOM hens were overall heavier than WT controls but this was statistically significant only between day 8 and day 24 ($P < 0.05$). **C:** Sums of histopathological scores for IL10KO WT ($n=5$) and HOM ($n=9$), and for IL10EnKO WT ($n=9$) and HOM ($n=11$) birds raised in the NARF SPF facility; tissue samples were collected at regular intervals from 16 to 40 weeks post-hatch. No significant differences were observed between genotypes for the tissues analysed. **D:** Growth curves for IL10KO WT ($n=2$ to 10) and HOM ($n=5$ to 15) hens raised in the NARF conventional facility, from hatch to 48 weeks. IL10KO HOM hens were significantly smaller than WT controls from 3 to 19 weeks post-hatch ($P < 0.05$), but this difference resolved with age. **E:** Sums of histopathological scores for IL10KO WT and HOM birds ($n=11$ in each group) raised in the NARF conventional facility; tissue samples were collected at regular intervals from 10 to 50 weeks post-hatch. No significant differences were observed between genotypes for the tissues analysed. **F:** Cellular infiltration scores for IL10KO WT and HOM birds raised in the NARF SPF and conventional facilities (same bird numbers as in C and E). Cellular infiltration scores were overall significantly higher for IL10KO WT and HOM birds raised in the conventional facility. Maximum possible sums of scores for data in C and E = 11; score range for data in F = 0 to 3. Data displayed as mean with SD (A, B and D) or as median with 95% confidence interval (C, E and F). Statistical significance calculated using one-way ANOVA with Bonferroni multiple comparison tests (A-B), Kruskal-Wallis test (C), two-tailed unpaired t tests (D), or Mann-Whitney U tests (E, F); * $P < 0.05$, ** $P < 0.01$, ns: not significant.

Statistical analyses revealed differences in vaccine response between IL10KO WT and HOM birds. Antibody titres to AEV and IBDV were significantly lower in IL10KO HOM birds compared to WT controls, whereas those to EDS and IBV were significantly higher in IL10KO HOM birds compared to WT controls (SI Appendix, Figure S6). AEV and IBDV vaccination were carried out with live vaccines delivered in the drinking water, whereas EDS and the last dose of IBV vaccination were done with inactivated vaccines given by intramuscular injection (SI Appendix, Table S2). Antibody titres to CAV, ILTV and NDV were not significantly different between IL10KO HOM birds and WT controls (SI Appendix, Figure S6).

The impact of IL10 deficiency on the response to enteric bacterial infection

Campylobacter jejuni (*C. jejuni*) and non-typhoidal *Salmonella* are zoonotic diarrhoeal pathogens of global importance [33]. Breed-specific pathology associated with *C. jejuni* infection in chickens was associated with relatively low caecal *IL10* mRNA [34]. Similarly, IL10-deficient C57BL/6J mice exhibited severe inflammation in the caecum and colon upon *C. jejuni* infection, whereas congenic wild-type mice were persistently colonised in the absence of gut pathology [35]. A similar outcome was reported in IL10-deficient mice infected with *Salmonella* Typhimurium (*S. Typhimurium*) [36].

To study the role of IL10 during *Campylobacter* infection in chickens, two cohorts of 20 IL10KO WT, HET or HOM chickens obtained from the NARF SPF avian facility were inoculated at 14 days-old with 10^2 colony forming units (CFU) *C. jejuni* strain M1 and analysed 1- and 2-weeks post-infection. The dose of strain M1 used is the lowest needed to achieve reliable caecal colonisation [37]. In the first cohort, colonisation of the caeca by *C. jejuni* in IL10KO HOM birds was significantly lower at 1-week post-infection, with a median of $4.8 \log_{10}$ CFU g^{-1} compared to $>9 \log_{10}$ CFU g^{-1} in WT birds (Figure 4A). However, this was not sustained and by 2-weeks post-infection all groups had caecal colonisation at $>9 \log_{10}$ CFU g^{-1} (Figure 4A). In the second cohort, median counts were not statistically different between IL10KO WT and HOM birds at 1-week post-infection, but caecal colonisation decreased by $\sim 2 \log_{10}$ CFU g^{-1} in IL10KO HOM birds at 2-weeks post-infection relative to WT birds (Figure 4A). The IL10KO HET birds did not differ significantly from WT in either cohort (Figure 4A). Histopathological analysis revealed that the caecal mucosa of IL10KO HOM birds presents with significantly increased numbers of lymphocytes and plasma cells, admixed with infiltrating heterophils, when compared to WT and IL10KO HET birds in both cohorts (SI Appendix, Figures S7 and S8). To confirm the apparent hyperinflammatory response in IL10KO HOM birds, we used multiplex quantitative reverse transcriptase-PCR (qRT-PCR) to measure expression of immunity-related genes in the caecum, as described [38]. In both cohorts we detected large increases in expression of pro-inflammatory cytokines, chemokines and other effectors, notably *Nos2*, in the IL10KO HOM caecum relative to WT (Figure 4B).

To test the role of IL10 in response to *Salmonella* infection, groups of 20 IL10KO WT, HET or HOM chickens were inoculated at 14 days-old with 10^8 CFU *S. Typhimurium* strain ST4/74 nal^R [39]. The impact of lack of IL10 on bacterial burden in caecum, liver and spleen was marginal at 1 week, but by 2 weeks the majority of IL10KO HOM birds had eliminated the pathogen, whereas WT birds retained bacteria in both caecum and spleen (Figure 4C). Enhanced lymphoplasmacytic and heterophil infiltration was observed in the ileum and caecum of IL10KO HOM compared to WT or IL10KO HET birds at both intervals after *S. Typhimurium* infection (SI Appendix, Figure S9). As observed following the *C. jejuni* challenge, analysis of inflammatory gene expression by multiplex qRT-PCR revealed a substantial increase in response in the IL10KO HOM birds relative to WT (Figure 4D). Volcano plots showing the direction and magnitude of gene expression differences detected between IL10KO WT and HOM birds following bacterial infection are shown in SI Appendix, Figures S10 and S11. In summary, the absence of IL10 leads to a sustained increase in inflammation in response to enteric bacterial infection that is also associated with more effective clearance.

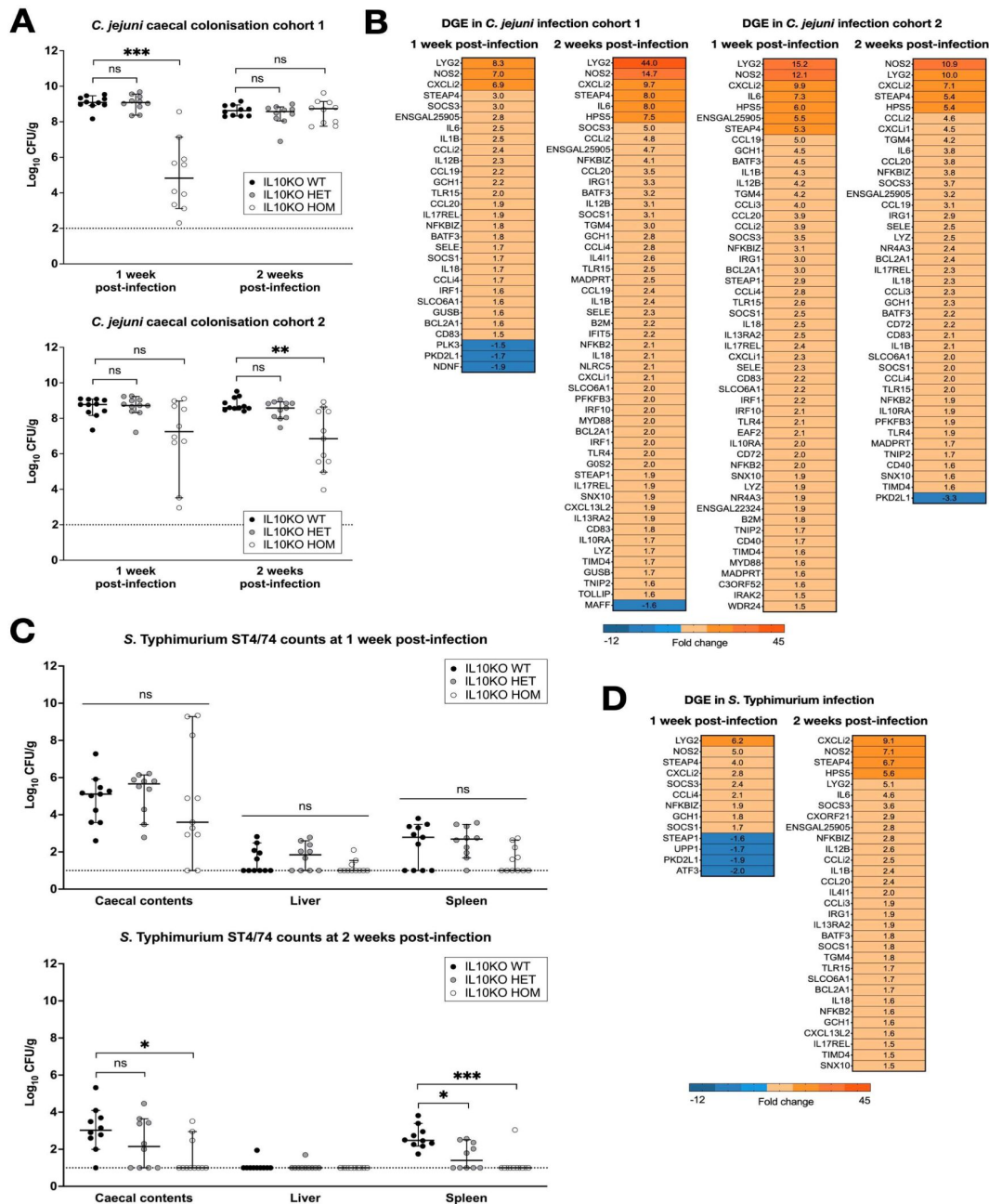


Figure 4

Net replication of *C. jejuni* and *S. Typhimurium* in the caeca of IL10KO WT, HET and HOM chickens and immune responses to infection.

A: Caecal burden of *C. jejuni* strain M1 in IL1KO WT, HET and HOM chickens at 1- and 2-weeks post-inoculation in two separate trials. **B:** Heat map of differentially transcribed genes measured by multiplex qRT-PCR in the caeca of IL10KO HOM chickens in relation to IL10KO WT birds at 1- and 2-weeks post-inoculation in two separate trials. An increase in the expression of pro-inflammatory genes was consistently observed in both studies. **C:** Burden of *S. Typhimurium* strain ST4/74 in the caeca, liver and spleen of IL10KO WT, HET and HOM chickens at 1- and 2-weeks post-inoculation (single trial). **D:** Heat map of differentially transcribed genes in the caeca of IL10KO WT and HOM chickens 1- and 2-weeks post-inoculation with ST4/74. All group sizes: n=10 or 11. Bacterial colonisation shown as median with 95% confidence interval; dotted line shows limit of detection for each study (A, C). Statistical significance calculated using one-way ANOVA with Kruskal-Wallis test followed by Dunn's multiple comparison tests; * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, ns: not significant.

The impact of IL10 deficiency on the outcome of enteric protozoan infection

Eimeria tenella (*E. tenella*) is one of ten protozoan parasite species that can cause coccidiosis in chickens [40]. *Eimeria* infection of chickens induces a robust IFN γ -driven immune response with evidence of higher IL10 expression associated with increased susceptibility [12, 41]. Following an initial titration, an infectious dose of 7000 sporulated *E. tenella* oocysts per bird was set as the standard challenge, as this induced measurable phenotypes in pathology, performance, and parasite replication in the absence of overt disease in WT birds. Oral challenge with this or a mock dose was administered to cohorts of 10 IL10KO HOM and IL10EnKO HOM chickens and corresponding WT and HET controls on day 21 post-hatch. The outcomes are shown in **Figure 5**. Average parasite replication was determined by quantitative PCR (qPCR) as the ratio of parasite genomes detected per host genome in total caecal tissue collected six days post-infection. Parasite burden was almost ablated in IL10KO HOM and significantly reduced in IL10EnKO HOM chickens compared to their respective WT controls, whereas HET mutations had no significant effect in either case (**Figure 5A-B**). Histology confirmed reduced parasite replication in IL10KO HOM chickens, with dense populations of late-stage gametocytes observable in WT caecal tissues and correspondingly very sparse numbers in IL10KO HOM chickens (**Figure 5C-D**). This selective reduction of parasite burden in IL10KO HOM chickens was associated with a significant increase in caecal lesion score and a substantial reduction in body weight gain over the 6-day challenge interval (**Figure 5E** and **G**). By contrast, both IL10KO HET and IL10EnKO HOM and HET (each anticipated to reduce IL10 expression) showed reduced caecal lesion scores and had no effect on body weight gain compared to their respective WT controls (**Figure 5E-H**).

To better characterise the differences in pathology, performance and parasite replication detected between IL10KO WT and HOM chickens, we undertook a time course study, analysing parasite burden by qPCR at 0-, 2-, 4-, 6-, 8- and 10-days post-infection. In parallel, excreted oocysts per gram (OPG) were counted in litter samples collected from IL10KO WT and HOM groups during the period of peak oocyst excretion at 7-, 8- and 9-days post-infection. **Figure 6A** confirms the greatly reduced *E. tenella* replication in IL10KO HOM, that was reflected in corresponding reduced excretion of oocysts (**Figure 6B**). Lesions associated with *E. tenella* infection were accelerated in IL10KO HOM chickens compared to WT (**Figure 6C**). The adverse impact of IL10KO HOM on body weight gain was sustained to day 8 and showed signs of recovery by day 10 (**Figure 6D**), likely due to complete pathogen elimination (**Figure 6A**). Consistent with the mild inflammatory cell infiltration in IL10KO HOM birds noted in **Figure 5D**, inflammatory gene expression was elevated in the caecum prior to challenge (Day 0). The greatest difference in IL10KO HOM birds relative to WT was detected on day 6, during the period of peak pathology (**Figure 6C** and **E**). Volcano plots showing the direction and magnitude of gene expression differences detected between IL10KO WT and HOM birds following *E. tenella* infection are shown in SI Appendix, Figure S12, highlighting genes associated with inflammation (*Il6*) and lysosomes (*Hps5*). Transcription of genes associated with movement of molecules across membranes (*Pkd2l1*, *Abcg2*) and tubulin production (*Tubat*) was lower in IL10KO HOM birds 6 days post-infection.

Discussion

An intrinsic compromise exists between effective host defence leading to pathogen elimination and the induction of immune-mediated pathology. The central importance of IL10 in mediating this balance has been studied extensively in the mouse, notably in the context of parasite infections [1, 4, 5, 11, 42]. The regulatory function of IL10 is especially important in preventing inappropriate responses to the intestinal microbiome with the potential to drive spontaneous enterocolitis [2, 8, 10, 27]. Here we have generated two *IL10* mutations in the chicken genome, a definitive knockout and a putative enhancer mutation. Notwithstanding the

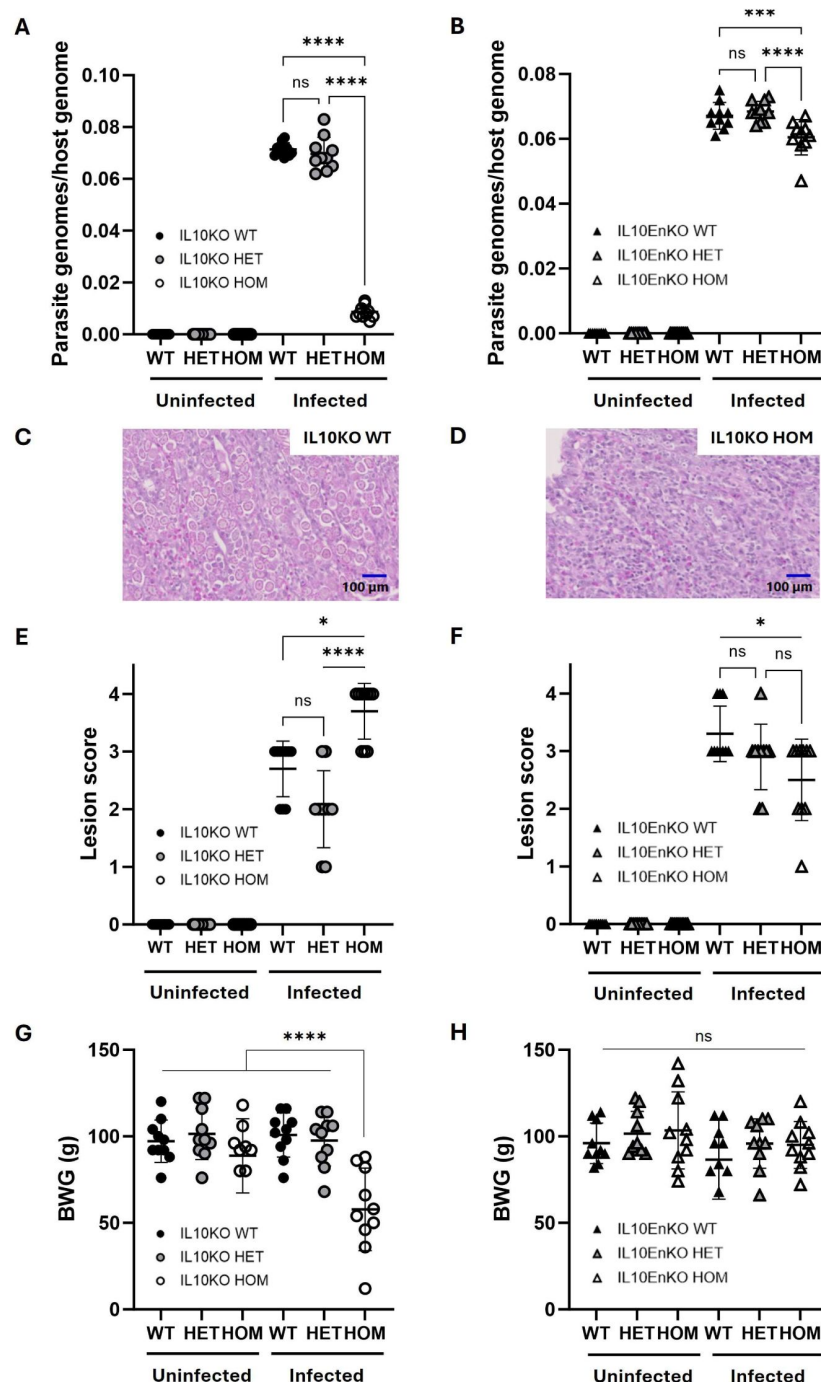


Figure 5

***Eimeria tenella* replication and impact on host pathology and performance in WT, HET and HOM G2 populations of IL10KO and IL10EnKO chickens.**

A-B: Parasite replication calculated as parasite genomes per host genome detected by quantitative PCR in genomic DNA extracted from caecal tissue six days post-infection (dpi). **C-D:** Haematoxylin and eosin stained caecal tissues collected from IL10KO WT and HOM chickens six days post *Eimeria tenella* infection. Numerous gametocytes are visible in the IL10KO WT line (C); gametocytes are largely absent in the IL10KO HOM line (D). **E-F:** Parasite-associated pathology measured as caecal lesion scores six dpi. **G-H:** Bodyweight gain (BWG) over six days from time of challenge to sampling. All group sizes: n=10. Statistical significance calculated using one-way ANOVA with Tukey's multiple comparison tests (A-B, G-H) or Kruskal-Wallis with Dunn's multiple comparison tests (E-F); * $P < 0.05$, *** $P < 0.001$, **** $P < 0.0001$, ns: not significant.

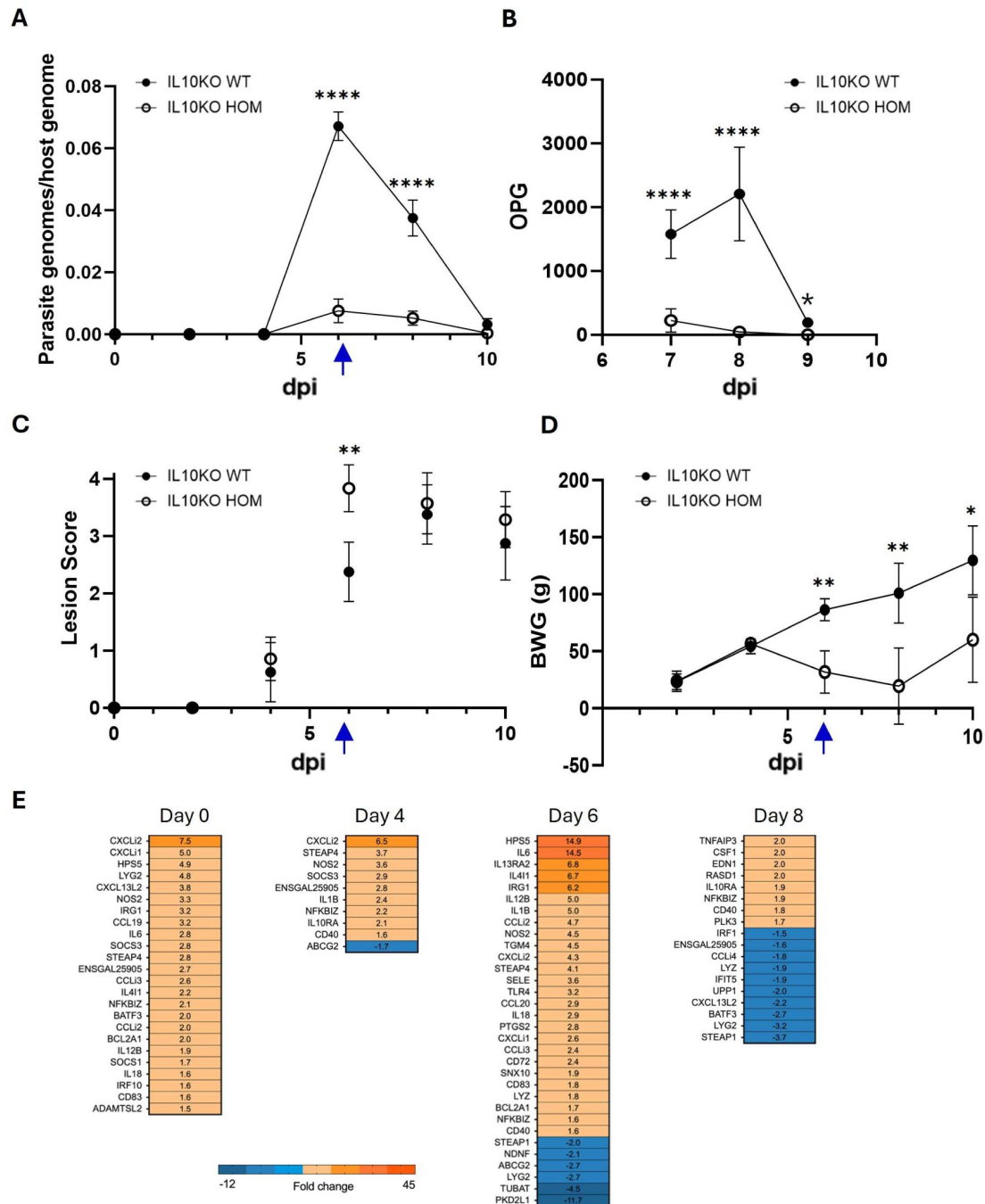


Figure 6

Eimeria tenella infection time course in IL10KO WT and HOM chickens.

A: Parasite replication calculated as parasite genomes per host genome detected by quantitative PCR in genomic DNA extracted from caecal tissue. **B:** Oocysts per gram (OPG) litter. **C:** Parasite-associated pathology measured as caecal lesion scores. **D:** Bodyweight gain (BWG) over ten days from time of challenge to final sampling. **E:** Heat map of differentially transcribed genes in the IL10KO HOM caeca relative to IL10KO WT chickens 0-, 4-, 6- and 8-days post-infection (dpi). 48 chicks per group, eight culled from each group per timepoint. Statistical significance calculated using mixed-effects model (REML) with Tukey's multiple comparison tests (A, D), two-way ANOVA with Šidák's multiple comparison tests (B), or Mann-Whitney U tests (C); * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$, ns: not significant. Blue arrows indicate the time of sampling in the comparative phenotyping study (Figure 5).

development of robust technologies based upon CRISPR/Cas9, the development of models of immune deficiency in chickens remains time consuming and logistically challenging, and relatively few informative mutations have been generated [20, 43]. The enhancer mutation was intended to generate a quantitative impact on IL10 expression to model selection for reduced expression in production lines of birds. Based upon the mouse *Il10*^{-/-} phenotype, we also anticipated a possible severe post-hatch failure to thrive due to spontaneous colitis in IL10KO HOM chicks. An unexpected outcome was the discovery that the IL10KO in chicken is not dosage compensated, therefore the HET provides a second model of the impact of reduced expression. In overview, neither the IL10KO HET nor the IL10EnKO HOM had a large effect on pathogen clearance or inflammation in either of the challenge models, indicating that IL10 biology is not highly sensitive to changes in expression of the magnitude that has been detected in individual lines of commercial broilers and layers [14, 44, 45].

Detailed analysis of the IL10KO HOM birds in SPF or conventional avian facilities revealed no evidence of spontaneous intestinal inflammation or compromised body weight gain, and only a small and inconsistent effect on specific antibody responses to routine live or inactivated vaccines. Since spontaneous colitis in mice depends upon resident enteric bacteria [3], it is possible that the difference compared to the rodent models lies in the nature of the chicken intestinal microbiome [46], in which case the changes that we observed in the microbiome of IL10KO HOM chicks may serve to mitigate potential spontaneous intestinal pathology. In any case, the results indicate that deletion of IL10 from the genome or selection in favour of low IL10 expression alleles [14, 17, 44, 45] need not necessarily affect production traits.

In both the bacterial challenge models, and in the *Eimeria* challenge, the response of IL10KO HOM bird highlights for the first time in a non-rodent species the compromise between pathogen elimination and pathology. Accelerated pathogen clearance is associated with increased inflammatory gene expression and local inflammation, and in the case of *Eimeria*, reduced body weight gain. The models we have chosen are not lethal and there is effective pathogen clearance even in WT birds. Findings with *Campylobacter* in IL10KO HOM birds are consistent with elevated intestinal inflammation in a broiler breed with relatively low IL10 expression, albeit a significant reduction in caecal colonisation relative to breeds with higher IL10 expression was not observed [34]. While the caecal load of *C. jejuni* was markedly reduced in IL10KO HOM birds at most intervals examined, some inconsistency existed in the timepoints when the reduction reached statistical significance between cohorts, likely as a consequence of inter-animal variation. This is typical of *C. jejuni* colonisation in chickens, where bacterial population structures have been reported to be variable and unpredictable [47]. Similar variation between time intervals, birds and repeated experiments has been reported when evaluating vaccines against *C. jejuni* colonisation (e.g. [48, 49]). In mammals, *Salmonella* has been reported to gain a competitive advantage from intestinal inflammation, both owing to availability of tetrathionate that *Salmonella* can uniquely use as a respiratory electron acceptor [50] and remodelling of the indigenous microbiota [51]. It is unclear if *Salmonella* derives the same benefits during colonisation of the avian intestine and the counts of viable bacteria recovered from the caeca of WT and IL10KO HOM birds can only be interpreted as the balance of bacterial growth and killing.

Our results with *Eimeria* are relevant to commercial studies aimed at targeting IL10 to mitigate disease. Coccidiosis has been estimated to exert costs in excess of £10 billion every year to the chicken production industry [52], with *E. tenella* among the most common and pathogenic species [53, 54]. Feeding neutralizing anti-IL10 antibody to chickens during *Eimeria* infection was reported to mitigate reduced body weight gain (BWG) but had an inconsistent impact on parasite replication and pathology [55, 56]. Similarly, feeding antibody to IL10 receptor 2 (but not receptor 1) reduced lost BWG, but had no effect on oocyst shedding [57]. Feeding chickens genetically modified corn (*Zea mays*) expressing IL10 improved BWG and reduced oocyst shedding and caecal lesions [58]. These results are consistent with the phenotypes associated with a reduction, rather than complete absence, of IL10 observed here.

Given the excessive inflammatory activation we observed in non-lethal infections, and the literature in mice, IL10KO HOM birds or hypomorphs might be hyper-sensitive to more virulent bacterial infections and viral pathogens, such as avian influenza, that can elicit a cytokine storm. On the other hand, they may be primed for a more rapid and effective response. These possibilities can now be tested using the models we have developed.

Materials and methods

Animals

All chicken lines used in this study were bred and maintained at the National Avian Research Facility (NARF), University of Edinburgh, under UK Home Office Establishment Licence and in compliance with the Animals (Scientific) Procedures Act (ASPA) 1986. All regulated procedures were approved by local ethical review committees and carried out by UK Home Office Personal Licence holders under UK Home Office Project Licences 70/8528 and PP9565661 (creation and maintenance of genetically altered chicken lines), PCD70CB48 (*Campylobacter* and *Salmonella* challenges performed at the Moredun Research Institute) and PDAAD5C9D (*Eimeria* challenges performed at the Royal Veterinary College). Animals were humanely culled in accordance with Schedule 1, ASPA 1986.

The Rhode Island Red (RIR) line was maintained as a closed outbred population in the NARF specified pathogen-free (SPF) avian facility. The CSF1R-eGFP line [59] was created and maintained on a Hy-Line background in the NARF conventional avian facility by HOM x HOM crosses; CSF1R-eGFP fertile eggs were used for PGC derivation. The iCaspase9 line [23] was created and maintained on a RIR background in the NARF SPF avian facility by crossing iCaspase9 HET cockerels to RIR hens; iCaspase9 fertile eggs were used as hosts for PGC injection. The IL10KO and IL10EnKO lines were created in the NARF SPF avian facility and maintained on a mixed Hy-Line / RIR background by crossing iCaspase9 surrogate host cockerels carrying gene-edited PGCs (generation G0) to RIR hens to produce the first generation (G1) of IL10KO HET and IL10EnKO HET chickens (see SI Appendix, Table S1 for detailed G1 bird numbers); the second generation (G2) of IL10KO and IL10EnKO chickens was obtained by crossing G1 HET cockerels to G1 HET hens from each line, resulting in WT, HET and HOM offspring (see SI Appendix, Table S1 for detailed G2 bird numbers).

Chicken PGC derivation and culture

PGC lines were derived from blood collected from CSF1R-eGFP HOM embryos at Hamburger-Hamilton (HH) stage 16 and expanded in FAOT medium as previously described [60]. A W-chromosome-specific PCR [61] was used to determine the sex of the PGC lines. PGC lines were tested for *Mycoplasma* contamination by PCR using primers specific for *Mycoplasma synoviae* (MS; Myco5_14713A and Myco3_14709) and *Mycoplasma gallisepticum* (MG; Myco5_14712 and Myco3_14709); see SI Appendix, Table S3 for primer sequences.

CRISPR plasmid, guide RNAs and ssODN donor

Guide RNAs (gRNAs) were designed using CHOPCHOP [62] and CRISPOR [63] and cloned into the High Fidelity CRISPR/Cas9 plasmid HF-PX459 V2.0 [64] as described previously [65]. A gRNA (ggIL10_exon1_g3) and a 143-bp single-stranded oligonucleotide DNA (ssODN; IL10_exon1_HDRoligo2; IDT Ultramer™ DNA Oligonucleotide) containing a STOP codon (TAG) and AvrII restriction site (CCTAGG), were used to edit *IL10* exon 1. Two gRNAs (IL10_enhancer_guide1 and IL10_enhancer_guide2) were used in combination to delete the *IL10* putative enhancer region. See SI Appendix, Table S4 for gRNA and ssODN sequences.

PGC transfection and screening

To edit *IL10* exon 1, 200,000 CSF1R-eGFP male PGCs were transiently transfected with 1.5 µg gRNA plasmid ggIL10_exon1_g3 and 10 pmol ssODN IL10_exon1_HDRoligo2. To delete the *IL10* putative enhancer region, 200,000 CSF1R-eGFP male PGCs were transiently transfected with 1.0 µg gRNA plasmid IL10_enhancer_guide1 and 1.0 µg gRNA plasmid IL10_enhancer_guide2. Transient transfections were performed with Lipofectamine 2000 (Invitrogen), followed by treatment with 0.4 µg/mL puromycin to enrich for transfected PGCs, as previously described [64]. Bulk PGC populations were screened by PCR to confirm the presence of the desired IL10KO or IL10EnKO edits. Single-cell clonal populations were subsequently expanded as previously described [64] and screened by PCR and sequencing. Primers IL10_Exon1_F4 and IL10_Exon1_R4 were used to amplify a 1826-bp fragment encompassing *IL10* exon 1, followed by digestion with AvrII to identify PGCs carrying the IL10KO edited allele (SI Appendix, Figure S4A); primers IL10_Exon1_F1 and IL10_Exon1_R1 were subsequently used to sequence the *IL10* exon 1 fragment. Primers IL10-Enhancer_F2 and IL10_Enhancer_R2 were used to amplify and sequence a fragment encompassing the *IL10* putative enhancer region (SI Appendix, Figure S4C). See SI Appendix, Table S3 for primer sequences and expected fragment sizes. PGCs were transfected and cultured under SPF conditions and regularly tested for mycoplasma contamination, as described above.

Generation of surrogate host chickens and establishment of the IL10KO and IL10EnKO lines under SPF conditions

Male PGCs carrying the desired IL10KO or IL10EnKO biallelic edits were injected into iCaspase9 embryos at HH stage 16, together with the iCaspase 9 activator AP20187 (Takara Bio) as previously described [23], to create surrogate host cockerels carrying IL10KO HOM or IL10EnKO HOM germ cells (G0 founders); all manipulations were performed under SPF conditions in the NARF SPF avian facility. G0 founders were hatched and raised to sexual maturity in negative pressure isolators and released to floor pens at 22 weeks. The SPF status of the G0 founders was externally assessed at three timepoints (10-, 20- and 30-weeks post-hatch) by screening for 17 different pathogens (Sci-Tech Ireland; SI Appendix, Table S5).

G0 founder cockerels were bred to RIR hens to produce the first generation (G1) of IL10KO HET and IL10EnKO HET chickens (see SI Appendix, Table S1 for detailed G1 bird numbers); the second generation (G2) of IL10KO and IL10EnKO chickens was obtained by crossing G1 HET cockerels to G1 HET hens, resulting in IL10KO and IL10EnKO WT, HET and HOM offspring (see SI Appendix, Table S1 for detailed G2 bird numbers). All G1 and G2 chicks were genotyped by PCR using genomic DNA extracted from chorioallantoic membrane or 4-µL blood samples as previously described [23]. Primers IL10_Exon1_F1 and IL10_Exon1_R1 were used to amplify a 246-bp fragment encompassing *IL10* exon 1, followed by digestion with AvrII to identify the IL10KO WT and edited alleles (SI Appendix, Figure S4B). Primers IL10-Enhancer_F2 and IL10_Enhancer_R2 were used to amplify the IL10EnKO WT and edited alleles (SI Appendix, Figure S4D). Primers LTR_U3_F, 235_F and 235_R were used to amplify the CSF1R-eGFP WT and transgenic alleles. See SI Appendix, Table S3 for primer sequences and expected fragment sizes. The SPF status of G1 and G2 chickens was verified at point of lay (20-24 weeks) and at end of life (around 40 weeks).

Animal monitoring, post-mortem examination and histopathological analyses under SPF and conventional conditions

All IL10KO and IL10EnKO chicks hatched in the NARF SPF avian facility were closely monitored for signs of reduced growth, health issues, adverse phenotypes or unexpected behaviours. All chicks were weighed daily in the first week post-hatch and twice a week until four weeks; a small cohort of G2 IL10KO and IL10EnKO chicks was subsequently weighed weekly, up to a maximum of 30 weeks. Post-mortem examination of one WT and one HOM bird from both IL10KO and IL10EnKO lines was performed monthly from 16 to 40 weeks, with a particular focus on the

gastrointestinal tract. Tissue samples from the proximal colon, ileum and duodenum were collected for histopathological analyses at all timepoints and scored in a blinded manner using a semiquantitative scoring system [25] to grade the extent of destruction of normal mucosal architecture (0-3), the presence and degree of cellular infiltration (0-3), the extent of muscle thickening (0-3), the presence or absence of crypt abscesses (0-1), and the presence or absence of goblet cell depletion (0-1), for a maximum possible score of 11.

In addition to animals raised in the NARF SPF avian facility, a single cohort of IL10KO chicks was hatched in the NARF conventional avian facility and closely monitored as described above. Post-mortem examination of one WT and one HOM IL10KO birds was performed monthly from 10 to 50 weeks, with a particular focus on the gastrointestinal tract. Tissue samples from the proximal colon, duodenum and caecum were collected for histopathological analyses and scored as described above.

Generation of bone marrow-derived macrophage from chicken embryos

Bone marrow cells were isolated from the tibias and femurs of Day 18 embryos (by pooling three embryos of the same genotype per sample) by flushing the bone marrow through a 40 μ m cell strainer with RPMI-1640 medium (Sigma R5886) supplemented with 10% fetal bovine serum ultra-low IgG (Gibco 011-90035M), 2 mM L-glutamine (Gibco 25030081) and 0.1X penicillin/streptomycin (Gibco 15140122) using a syringe and blunt needle. Bone marrow cells were washed, resuspended at 1×10^6 cells/mL in supplemented RPMI-1640 medium, seeded at 1 mL per well in 6-well plates or 150 μ L per well in 96-well plates, and cultured for 7 days at 41°C, 5% CO₂ in the presence of 200 ng/mL colony stimulating factor-1 (CSF-1) to induce formation of bone marrow-derived macrophages (BMDM). Cell culture supernatants were subsequently used for capture ELISA and nitric oxide assay, as detailed below.

Detection of IL10 by capture ELISA

On day 7 of culture, BMDMs cultured in 6-well plates were stimulated with 0.5 μ g/mL lipopolysaccharide (LPS from *E. coli* O55:B5; Sigma L2880) for 2 hours at 41°C, 5% CO₂ to induce IL10 expression; supernatants were then harvested and IL10 protein levels were measured by capture ELISA, as described previously [24]. Briefly, assay plates were coated with 3 μ g/mL capture antibody (ROS-AV164; [24]) incubated overnight at 4°C, washed and blocked. Plates were then incubated with recombinant IL10 standards or 50 μ L of 10-fold diluted cell culture supernatants, followed by incubation with 1 μ g/mL biotinylated detection antibody (ROS-AV163; [24]) and then incubation with PierceTM High Sensitivity Streptavidin-HRP (1:5000; Thermo Fisher Scientific, cat #34028), before adding 1-StepTM TMB ELISA Substrate Solution (Thermo Fisher Scientific, cat #21130) and then sulfuric acid stop solution. Absorbance was measured at 450 nm (550 nm as a reference) in a SpectraMax 250 microplate spectrophotometer system (Molecular Devices, Sunnyvale, CA, USA). Data were fitted with GraphPad Prism 7.0 using a second-order polynomial (quadratic) model; statistical significance was calculated using two-tailed unpaired t tests.

Nitric oxide assay

On day 7 of culture, BMDMs cultured in 96-well plates were stimulated with 0.5 μ g/mL LPS in the presence or absence of 2.5 μ g/mL neutralizing anti-IL10 antibody (ROS-AV163; [24]) for 24 hours at 41°C, 5% CO₂; supernatants were then harvested and nitrite levels were measured by Griess assay, according to the manufacturer's instructions (Promega G2930); absorbance was measured at 550 nm. Statistical significance was calculated using two-tailed unpaired t tests.

Response to vaccination

IL10KO WT and HOM chicks hatched and raised in the NARF conventional avian facility were vaccinated following the routine vaccination schedule in place in the facility (SI Appendix, Table S2). One-mL blood samples were collected from the brachial vein of 29-weeks old hens and sent fresh to Sci-Tech Ireland for antibody titre measurements by ELISA. Statistical significance was calculated using Mann-Whitney U tests.

Microbiome analysis

Caecal contents were harvested from 4-weeks old IL10KO WT and HOM chicks raised in the NARF SPF avian facility. DNA isolation, amplification of the V3-V4 region of the 16S rDNA gene and pair-end sequencing were performed as described previously [32]. Raw reads were quality assessed using fastQC, and the DADA2 plugin in QIIME2 was used to generate an ASVs table against the SILVA138 database. Alpha diversity and weighted uniFrac distances were calculated using the Phyloseq package, while Bray-Curtis metrics were generated using the vegan package. PERMANOVA (Bray-Curtis) analysis was conducted to test for significant effects of the group on overall microbiome community composition. LEfSe analysis was performed to identify differentially abundant phyla and genera between both groups using a linear discriminant analysis (LDA) score equal to 4 as a threshold value. *P* values were declared significant at *P* < 0.05. The sequencing data have been deposited in the European Nucleotide Archive (ENA) at EMBL-EBI under accession number PRJEB71960.

Bacterial strains

Campylobacter jejuni strain M1 was used as the minimum number of viable bacteria required for reliable intestinal colonisation of chickens has been defined [37]. Strain M1 was grown on charcoal cefoperazone deoxycholate agar (CCDA) at 41 °C for 48 hours in a microaerophilic atmosphere (90 % N₂, 5 % CO₂ and 5 % O₂). For challenge, 10 mL of Mueller Hinton broth was inoculated with single colonies and incubated for 16 hours with shaking at 41 °C in the same atmosphere. Prior to inoculation, bacteria were confirmed to have spiral morphology and to be motile by phase contrast microscopy. Cultures were diluted in phosphate-buffered saline (PBS) based on a standard curve of colony-forming units (CFU/mL) relative to optical density at 600 nm and inocula were retrospectively confirmed by plating serial ten-fold dilutions onto CCDA. A spontaneous nalidixic acid-resistant mutant of *Salmonella enterica* serovar Typhimurium strain 4/74 was used, as its colonisation kinetics are well defined in chickens [39]. This was cultured on Luria Bertani (LB) agar containing 20 µg/mL nalidixic acid at 37 °C overnight and a single colony was transferred to LB broth with nalidixic acid and incubated at 37 °C with shaking for 20 hours. 100 µL of this culture was used for inoculation and the number of viable bacteria given was determined retrospectively by plating serial ten-fold dilutions onto LB agar.

Experimental infection of chickens

Animal experiments using *Campylobacter* or *Salmonella* were conducted at the Moredun Research Institute. Birds were provided with access to water and irradiated feed based on vegetable protein *ad libitum*. Groups of 20 to 22 birds of each genotype (IL10KO WT, HET, HOM) were wing-tagged for identification and separately housed in biosafety level 2 facilities from day 7 of age in colony cages. At 14 days of age, birds were inoculated by oral gavage with 100 µL of bacterial suspension. In two separate trials with *C. jejuni*, birds were inoculated with 1×10² CFU of strain M1. In a single trial with *S. Typhimurium*, birds were inoculated with 1×10⁸ CFU of strain 4/74 nal^R. At 7 days post-inoculation, half of the birds in each group were euthanised by cervical dislocation. The remainder of the birds were euthanised at 14 days post-inoculation. On post-mortem examination, samples from the ileum and the distal end of the caecum were stored in RNAlater® or fixed in 10% neutral-buffered formalin for histopathology. The contents of the caeca were pooled at the bird

level, homogenised in PBS, and ten-fold serial dilutions cultured on CCDA at 41 °C in a microaerophilic atmosphere to enumerate CFU/g of *C. jejuni* or MacConkey agar containing 20 µg/mL nalidixic acid at 37 °C aerobically to enumerate CFU/g of *Salmonella*.

Histopathology

Formalin-fixed tissue samples were embedded in paraffin, sectioned, mounted on glass slides, and stained by haematoxylin-eosin (HE) staining. Histopathological analyses were performed by a certified veterinary pathologist blind to the genotype of individuals and the inoculum. Samples of ileum and caecum were assessed for the presence of lymphoplasmacytic and heterophil infiltration on a scale of 0 to 5 (where infiltration is 0 normal, 1 slight, 2 mild, 3 moderate, 4 marked or 5 severe).

RNA extraction and multiplex PCR

Caecal tissue samples in RNeasy® were homogenised with a hand-held tissue homogeniser and RNA was extracted using RNeasy kit following the instructions from the manufacturer (QIAGEN). The concentration and integrity of the RNA were respectively analysed using a NanoDrop microvolume spectrophotometer and TapeStation automated electrophoresis system. Reverse transcription, preamplification and high-throughput quantitative PCR (qPCR) for avian immune-related transcripts and reference genes were performed as previously described using a 96.96 Dynamic Array IFC for Gene Expression ([38]; Standard BioTools, San Francisco, CA, United States).

Parasites

Animal experiments using *Eimeria* were conducted at the Royal Veterinary College, using chicks hatched at the NARF SPF avian facility. All chickens were provided with access to water and ammonia fumigated feed based on vegetable protein *ad libitum*. The *Eimeria tenella* Houghton (H) reference strain was used [66]. Parasites were passaged by oral inoculation of specified pathogen-free Lohmann Valo chickens, purified and prepared for use as described elsewhere [54]. At 21 days post-hatch, chicks were weighed and infected by oral inoculation with 7000 sporulated *E. tenella* oocysts or mock-infected using sterile water. Group sizes (minimum eight to ten per group) were determined following dose titration in IL10KO and IL10EnKO HET chickens using power calculations with alpha set at 0.05 and power at 80%. At the desired timepoint, birds were culled and whole blood was collected for serum. Caeca were collected, scored for severity of infection [67] and preserved using RNeasy (Invitrogen). Total genomic DNA (gDNA) was extracted from thawed complete caeca stored in RNeasy from all studies. Caeca were homogenised in Buffer ATL using a TissueRuptor homogenizer (Qiagen, Hilden, Germany) and then digested overnight at 56 °C in Buffer ATL and proteinase K, prior to extraction using the Qiagen DNeasy Blood and Tissue DNA Kit according to manufacturer's instructions.

Quantitative PCR for assessment of *E. tenella* genome copy number was performed as previously described [68]. Briefly, gDNA extracted from caecal tissue was used as template for qPCR targeting *E. tenella* (RAPD-SCAR marker Tn-E03-116, primers F: 5'-TCGTCCTTGGCTGGCTATTC-3', R: 5'-CAGAGAGTCGCCGTCACAGT-3'), normalised against the number of chicken tata-binding protein (TBP) genomic copies detected (F: 5'-TAGCCCGATGATGCCGAT-3', R: 5'-GTTCCCTGTGTCGCTTGC-3'). Quantitative PCR was performed in 20 µL reactions in triplicate containing 10 µL 2 × SsoFast EvaGreen Supermix (Bio-Rad, Hercules, CA, USA), 1 µL of primers (F: 3 µM, R: 3 µM), 8 µL of molecular biological grade water (Invitrogen) and 1 µL of gDNA, or water as negative control. Hard-shelled 96-well reaction plates (Bio-Rad) were sealed with adhesive film (Bio-Rad) and loaded into a Bio-Rad CFX qPCR cycler. Reactions were heated to 95°C for 2 min, prior to 40 cycles consisting of 95°C for 15 s then 60°C for 30 s with a fluorescence reading taken after each cycle. Melting curve analysis was performed for 15 s at 95°C, before cooling to 65°C for 60 s, then heating to 95°C in 0.5°C increments for 0.5 s. Absolute quantification was performed against a standard

curve generated using serially diluted plasmid DNA containing the amplicon of interest (EtenSCAR or ChickenTBP), to generate a standard curve ranging from 10^6 to 10^1 genome copies per mL. Parasite genome copy number was normalised by division with host (chicken) genome copy number.

Statistical analyses

Minitab Statistical Software was used to analyse IL10 protein levels and nitric oxide production in BMDMs using two-tailed unpaired t tests, growth curves were analysed using two-tailed unpaired t tests or one-way ANOVA with Bonferroni multiple comparison tests, and histopathological scores were analysed using Mann-Whitney U tests or Kruskal-Wallis tests. Analysis of bacterial counts, histopathology, and heatmaps was performed with GraphPad Prism software version 10.2.0. Data from multiplex RT-qPCR was analysed with GenEx6 software. The distribution of data was analysed by D'Agostino-Pearson and Shapiro-Wilk normality tests. *Eimeria* replication and chicken body weight gain during parasite infection were analysed using one-way ANOVA with Tukey's multiple comparison tests, and differences in lesion scores were tested by Kruskal-Wallis with Dunn's multiple comparison tests. Bacterial counts and histopathology scores were analysed by Mann-Whitney U tests or Kruskal-Wallis tests. Fold change in transcript abundance was analysed by t tests using normalised data.

Data availability statement

All study data are included in the article and/or SI Appendix. IL10KO and IL10EnKO edited PGCs (HET and HOM) are cryopreserved and available to researchers under MTA.

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Author Contributions

Designed research: AP, DPB, DAH, DM, FT, KB, KW, MJM, MPS, RCT

Performed research: AAA, DPB, DM, GSA, JJO, KB, LG, LT, MV, RCT, ZW

Contributed new reagents/analytic tools: MJM

Analysed data: AAA, DB, DPB, DAH, DM, JDP, LG, LV, MJM, MPS, RCT, ZW

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Additional files

Supporting Information Appendix [🔗](#)

References

1. Saraiva M, Vieira P, O'Garra A (2020) **Biology and therapeutic potential of interleukin-10** *J Exp Med* **217** <https://doi.org/10.1084/jem.20190418> | PubMed | Google Scholar
2. Kuhn R, Lohler J, Rennick D, Rajewsky K, Muller W (1993) **Interleukin-10-deficient mice develop chronic enterocolitis** *Cell* **75**:263–74 [https://doi.org/10.1016/0092-8674\(93\)80068-p](https://doi.org/10.1016/0092-8674(93)80068-p) | PubMed | Google Scholar
3. Sellon RK, Tonkonogy S, Schultz M, Dieleman LA, Grenther W, Balish E, et al. (1998) **Resident enteric bacteria are necessary for development of spontaneous colitis and immune system activation in interleukin-10-deficient mice** *Infect Immun* **66**:5224–31 <https://doi.org/10.1128/IAI.66.11.5224-5231.1998> | PubMed | Google Scholar
4. Couper KN, Blount DG, Riley EM (2008) **IL-10: the master regulator of immunity to infection** *J Immunol* **180**:5771–7 <https://doi.org/10.4049/jimmunol.180.9.5771> | PubMed | Google Scholar
5. Rasquinha MT, Sur M, Lasrado N, Reddy J (2021) **IL-10 as a Th2 Cytokine: Differences Between Mice and Humans** *J Immunol* **207**:2205–15 <https://doi.org/10.4049/jimmunol.2100565> | PubMed | Google Scholar
6. Brockmann L, Soukou S, Steglich B, Czarnewski P, Zhao L, Wende S, et al. (2018) **Molecular and functional heterogeneity of IL-10-producing CD4(+) T cells** *Nat Commun* **9**:5457 <https://doi.org/10.1038/s41467-018-07581-4> | PubMed | Google Scholar
7. Pils MC, Bleich A, Prinz I, Fasnacht N, Bollati-Fogolin M, Schippers A, et al. (2011) **Commensal gut flora reduces susceptibility to experimentally induced colitis via T-cell-derived interleukin-10** *Inflamm Bowel Dis* **17**:2038–46 <https://doi.org/10.1002/ibd.21587> | PubMed | Google Scholar
8. Bernshtein B, Curato C, Ioannou M, Thaïss CA, Gross-Vered M, Kolesnikov M, et al. (2019) **IL-23-producing IL-10 α -deficient gut macrophages elicit an IL-22-driven proinflammatory epithelial cell response** *Sci Immunol* **4** <https://doi.org/10.1126/sciimmunol.aau6571> | PubMed | Google Scholar
9. Redhu NS, Bakthavatchalu V, Conaway EA, Shouval DS, Tsou A, Goettel JA, et al. (2017) **Macrophage dysfunction initiates colitis during weaning of infant mice lacking the interleukin-10 receptor** *eLife* **6** <https://doi.org/10.7554/eLife.27652> | PubMed | Google Scholar
10. Zigmond E, Bernshtein B, Friedlander G, Walker CR, Yona S, Kim KW, et al. (2014) **Macrophage-restricted interleukin-10 receptor deficiency, but not IL-10 deficiency, causes severe spontaneous colitis** *Immunity* **40**:720–33 <https://doi.org/10.1016/j.immuni.2014.03.012> | PubMed | Google Scholar

11. Piazzon MC, Lutfalla G, Forlenza M (2016) **IL10, A Tale of an Evolutionarily Conserved Cytokine across Vertebrates** *Crit Rev Immunol* **36**:99–129 <https://doi.org/10.1615/CritRevImmunol.2016017480> | PubMed | Google Scholar
12. Rothwell L, Young JR, Zoorob R, Whittaker CA, Hesketh P, Archer A, et al. (2004) **Cloning and characterization of chicken IL-10 and its role in the immune response to Eimeria maxima** *J Immunol* **173**:2675–82 <https://doi.org/10.4049/jimmunol.173.4.2675> | PubMed | Google Scholar
13. Russell KM, Smith J, Bremner A, Chintoan-Uta C, Vervelde L, Psifidi A, et al. (2021) **Transcriptomic analysis of caecal tissue in inbred chicken lines that exhibit heritable differences in resistance to Campylobacter jejuni** *BMC Genomics* **22**:411 <https://doi.org/10.1186/s12864-021-07748-2> | PubMed | Google Scholar
14. Boulton K, Nolan MJ, Wu Z, Psifidi A, Riggio V, Harman K, et al. (2018) **Phenotypic and genetic variation in the response of chickens to Eimeria tenella induced coccidiosis** *Genet Sel Evol* **50**:63 <https://doi.org/10.1186/s12711-018-0433-7> | PubMed | Google Scholar
15. Swaggerty CL, Pevzner IY, Kogut MH (2015) **Selection for pro-inflammatory mediators produces chickens more resistant to Eimeria tenella** *Poult Sci* **94**:37–42 <https://doi.org/10.3382/ps/peu053> | PubMed | Google Scholar
16. Calenge F, Lecerf F, Demars J, Feve K, Vignoles F, Pitel F, et al. (2009) **QTL for resistance to Salmonella carrier state confirmed in both experimental and commercial chicken lines** *Anim Genet* **40**:590–7 <https://doi.org/10.1111/j.1365-2052.2009.01884.x> | PubMed | Google Scholar
17. Ghebremicael SB, Hasenstein JR, Lamont SJ (2008) **Association of interleukin-10 cluster genes and Salmonella response in the chicken** *Poult Sci* **87**:22–6 <https://doi.org/10.3382/ps.2007-00259> | PubMed | Google Scholar
18. Ballantyne M, Taylor L, Hu T, Meunier D, Nandi S, Sherman A, et al. (2021) **Avian Primordial Germ Cells Are Bipotent for Male or Female Gametogenesis** *Front Cell Dev Biol* **9** <https://doi.org/10.3389/fcell.2021.726827> | PubMed | Google Scholar
19. Ioannidis J, Taylor G, Zhao D, Liu L, Idoko-Akoh A, Gong D, et al. (2021) **Primary sex determination in birds depends on DMRT1 dosage, but gonadal sex does not determine adult secondary sex characteristics** *Proc Natl Acad Sci U S A* **118** <https://doi.org/10.1073/pnas.2020909118> | PubMed | Google Scholar
20. Wu Z, Shih B, Macdonald J, Meunier D, Hogan K, Chintoan-Uta C, et al. (2023) **Development and function of chicken XCR1(+) conventional dendritic cells** *Front Immunol* **14** <https://doi.org/10.3389/fimmu.2023.1273661> | PubMed | Google Scholar
21. Idoko-Akoh A, McGrew MJ (2023) **Generation of Genome-Edited Chicken Through Targeting of Primordial Germ Cells** *Methods Mol Biol* **2631**:419–41 https://doi.org/10.1007/978-1-0716-2990-1_20 | PubMed | Google Scholar
22. Hume DA, Wollscheid-Lengeling E, Rojo R, Pridans C (2017) **The evolution of the macrophage-specific enhancer (Fms intronic regulatory element) within the CSF1R locus of vertebrates** *Sci Rep* **7**:17115 <https://doi.org/10.1038/s41598-017-15999-x> | PubMed | Google Scholar

23. Ballantyne M, Woodcock M, Doddamani D, Hu T, Taylor L, Hawken RJ, et al. (2021) **Direct allele introgression into pure chicken breeds using Sire Dam Surrogate (SDS) mating** *Nat Commun* **12**:659 <https://doi.org/10.1038/s41467-020-20812-x> | PubMed | Google Scholar
24. Wu Z, Hu T, Rothwell L, Vervelde L, Kaiser P, Boulton K, et al. (2016) **Analysis of the function of IL-10 in chickens using specific neutralising antibodies and a sensitive capture ELISA** *Dev Comp Immunol* **63**:206–12 <https://doi.org/10.1016/j.dci.2016.04.016> | PubMed | Google Scholar
25. McCafferty DM, Sihota E, Muscara M, Wallace JL, Sharkey KA, Kubes P (2000) **Spontaneously developing chronic colitis in IL-10/iNOS double-deficient mice** *Am J Physiol Gastrointest Liver Physiol* **279**:G90–9 <https://doi.org/10.1152/ajpgi.2000.279.1.G90> | PubMed | Google Scholar
26. Papoutsopoulou S, Pollock L, Walker C, Tench W, Samad SS, Bergey F, et al. (2021) **Impact of Interleukin 10 Deficiency on Intestinal Epithelium Responses to Inflammatory Signals** *Front Immunol* **12** <https://doi.org/10.3389/fimmu.2021.690817> | PubMed | Google Scholar
27. Buchler G, Wos-Oxley ML, Smoczek A, Zschemisch NH, Neumann D, Pieper DH, et al. (2012) **Strain-specific colitis susceptibility in IL10-deficient mice depends on complex gut microbiota-host interactions** *Inflamm Bowel Dis* **18**:943–54 <https://doi.org/10.1002/ibd.21895> | PubMed | Google Scholar
28. Devkota S, Wang Y, Musch MW, Leone V, Fehlner-Peach H, Nadimpalli A, et al. (2012) **Dietary-fat-induced taurocholic acid promotes pathobiont expansion and colitis in IL10^{-/-} mice** *Nature* **487**:104–8 <https://doi.org/10.1038/nature11225> | PubMed | Google Scholar
29. Gunasekera DC, Ma J, Vacharathit V, Shah P, Ramakrishnan A, Uprety P, et al. (2020) **The development of colitis in IL10^{-/-} mice is dependent on IL-22** *Mucosal Immunol* **13**:493–506 <https://doi.org/10.1038/s41385-019-0252-3> | PubMed | Google Scholar
30. Keubler LM, Buettner M, Hager C, Bleich A (2015) **A Multihit Model: Colitis Lessons from the Interleukin-10-deficient Mouse** *Inflamm Bowel Dis* **21**:1967–75 <https://doi.org/10.1097/MIB.0000000000000468> | PubMed | Google Scholar
31. Pandit RJ, Hinsu AT, Patel NV, Koringa PG, Jakhesara SJ, Thakkar JR, et al. (2018) **Microbial diversity and community composition of caecal microbiota in commercial and indigenous Indian chickens determined using 16s rDNA amplicon sequencing** *Microbiome* **6**:115 <https://doi.org/10.1186/s40168-018-0501-9> | PubMed | Google Scholar
32. Salavati Schmitz S, Salgado JP, Glendinning L (2024) **Microbiota of healthy dogs demonstrate a significant decrease in richness and changes in specific bacterial groups in response to supplementation with resistant starch, but not psyllium or methylcellulose, in a randomized cross-over trial** *Access Microbiol* **6** <https://doi.org/10.1099/acmi.0.000774.v4> | PubMed | Google Scholar
33. Havelaar AH, Kirk MD, Torgerson PR, Gibb HJ, Hald T, Lake RJ, et al. (2015) **World Health Organization Global Estimates and Regional Comparisons of the Burden of Foodborne Disease in 2010** *PLoS Med* **12**:e1001923 <https://doi.org/10.1371/journal.pmed.1001923> | PubMed | Google Scholar
34. Humphrey S, Chaloner G, Kemmett K, Davidson N, Williams N, Kipar A, et al. (2014) **Campylobacter jejuni is not merely a commensal in commercial broiler chickens and**

affects bird welfare *mBio* **5**:e01364–14 <https://doi.org/10.1128/mBio.01364-14> | PubMed | Google Scholar

35. Mansfield LS, Bell JA, Wilson DL, Murphy AJ, Elsheikha HM, Rathinam VA, et al. (2007) **C57BL/6 and congenic interleukin-10-deficient mice can serve as models of Campylobacter jejuni colonization and enteritis** *Infect Immun* **75**:1099–115 <https://doi.org/10.1128/IAI.00833-06> | PubMed | Google Scholar
36. Schultz BM, Salazar GA, Paduro CA, Pardo-Roa C, Pizarro DP, Salazar-Echegarai FJ, et al. (2018) **Persistent Salmonella enterica serovar Typhimurium Infection Increases the Susceptibility of Mice to Develop Intestinal Inflammation** *Front Immunol* **9**:1166 <https://doi.org/10.3389/fimmu.2018.01166> | PubMed | Google Scholar
37. Vohra P, Chintoan-Uta C, Terra VS, Bremner A, Cuccui J, Wren BW, et al. (2020) **Evaluation of Glycosylated FlpA and SodB as Subunit Vaccines Against Campylobacter jejuni Colonisation in Chickens** *Vaccines (Basel)* **8** <https://doi.org/10.3390/vaccines8030520> | PubMed | Google Scholar
38. Borowska D, Kuo R, Bailey RA, Watson KA, Kaiser P, Vervelde L, et al. (2019) **Highly multiplexed quantitative PCR-based platform for evaluation of chicken immune responses** *PLoS One* **14**:e0225658 <https://doi.org/10.1371/journal.pone.0225658> | PubMed | Google Scholar
39. Chaudhuri RR, Morgan E, Peters SE, Pleasance SJ, Hudson DL, Davies HM, et al. (2013) **Comprehensive assignment of roles for Salmonella typhimurium genes in intestinal colonization of food-producing animals** *PLoS Genet* **9**:e1003456 <https://doi.org/10.1371/journal.pgen.1003456> | PubMed | Google Scholar
40. Blake DP, Vrba V, Xia D, Jatau ID, Spiro S, Nolan MJ, et al. (2021) **Genetic and biological characterisation of three cryptic Eimeria operational taxonomic units that infect chickens (Gallus gallus domesticus)** *Int J Parasitol* **51**:621–34 <https://doi.org/10.1016/j.ijpara.2020.12.004> | PubMed | Google Scholar
41. Bremner A, Kim S, Morris KM, Nolan MJ, Borowska D, Wu Z, et al. (2021) **Kinetics of the Cellular and Transcriptomic Response to Eimeria maxima in Relatively Resistant and Susceptible Chicken Lines** *Front Immunol* **12** <https://doi.org/10.3389/fimmu.2021.653085> | PubMed | Google Scholar
42. Redpath SA, Fonseca NM, Perona-Wright G (2014) **Protection and pathology during parasite infection: IL-10 strikes the balance** *Parasite Immunol* **36**:233–52 <https://doi.org/10.1111/pim.12113> | PubMed | Google Scholar
43. von Heyl T, Klinger R, Aumann D, Zenner C, Alhussien M, Schlickerrieder A, et al. (2023) **Loss of alphabeta but not gammadelta T cells in chickens causes a severe phenotype** *Eur J Immunol* **53**:e2350503 <https://doi.org/10.1002/eji.202350503> | PubMed | Google Scholar
44. Boulton K, Nolan MJ, Wu Z, Riggio V, Matika O, Harman K, et al. (2018) **Dissecting the Genomic Architecture of Resistance to Eimeria maxima Parasitism in the Chicken** *Front Genet* **9** <https://doi.org/10.3389/fgene.2018.00528> | PubMed | Google Scholar
45. Freem L, Summers KM, Gheyas AA, Psifidi A, Boulton K, MacCallum A, et al. (2019) **Analysis of the Progeny of Sibling Matings Reveals Regulatory Variation Impacting the**

Transcriptome of Immune Cells in Commercial Chickens *Front Genet* **10**:1032 <https://doi.org/10.3389/fgene.2019.01032> | PubMed | Google Scholar

46. Bajagai YS, Van TTH, Joat N, Chousalkar K, Moore RJ, Stanley D (2024) **Layer chicken microbiota: a comprehensive analysis of spatial and temporal dynamics across all major gut sections** *J Anim Sci Biotechnol* **15**:20 <https://doi.org/10.1186/s40104-023-00979-1> | PubMed | Google Scholar
47. Coward C, van Diemen PM, Conlan AJ, Gog JR, Stevens MP, Jones MA, et al. (2008) **Competing isogenic Campylobacter strains exhibit variable population structures in vivo** *Appl Environ Microbiol* **74**:3857–67 <https://doi.org/10.1128/AEM.02835-07> | PubMed | Google Scholar
48. Buckley AM, Wang J, Hudson DL, Grant AJ, Jones MA, Maskell DJ, et al. (2010) **Evaluation of live-attenuated Salmonella vaccines expressing Campylobacter antigens for control of C. jejuni in poultry** *Vaccine* **28**:1094–105 <https://doi.org/10.1016/j.vaccine.2009.10.018> | PubMed | Google Scholar
49. Nothaft H, Perez-Munoz ME, Yang T, Murugan AVM, Miller M, Kolarich D, et al. (2021) **Improving Chicken Responses to Glycoconjugate Vaccination Against Campylobacter jejuni** *Front Microbiol* **12** <https://doi.org/10.3389/fmicb.2021.734526> | PubMed | Google Scholar
50. Winter SE, Thiennimitr P, Winter MG, Butler BP, Huseby DL, Crawford RW, et al. (2010) **Gut inflammation provides a respiratory electron acceptor for Salmonella** *Nature* **467**:426–9 <https://doi.org/10.1038/nature09415> | PubMed | Google Scholar
51. Rogers AWL, Tsolis RM, Baumler AJ (2021) **Salmonella versus the Microbiome** *Microbiol Mol Biol Rev* **85** <https://doi.org/10.1128/MMBR.00027-19> | PubMed | Google Scholar
52. Blake DP, Knox J, Dehaeck B, Huntington B, Rathinam T, Ravipati V, et al. (2020) **Re-calculating the cost of coccidiosis in chickens** *Vet Res* **51**:115 <https://doi.org/10.1186/s13567-020-00837-2> | PubMed | Google Scholar
53. Clark EL, Macdonald SE, Thenmozhi V, Kundu K, Garg R, Kumar S, et al. (2016) **Cryptic Eimeria genotypes are common across the southern but not northern hemisphere** *Int J Parasitol* **46**:537–44 <https://doi.org/10.1016/j.ijpara.2016.05.006> | PubMed | Google Scholar
54. Long PL, Millard BJ, Joyner LP, Norton CC (1976) **A guide to laboratory techniques used in the study and diagnosis of avian coccidiosis** *Folia Vet Lat* **6**:201–17 | PubMed | Google Scholar
55. Rasheed MSA, Tiwari UP, Jespersen JC, Bauer LL, Dilger RN (2020) **Effects of methylsulfonylmethane and neutralizing anti-IL-10 antibody supplementation during a mild Eimeria challenge infection in broiler chickens** *Poult Sci* **99**:6559–68 <https://doi.org/10.1016/j.psj.2020.09.046> | PubMed | Google Scholar
56. Sand JM, Arendt MK, Repasy A, Deniz G, Cook ME (2016) **Oral antibody to interleukin-10 reduces growth rate depression due to Eimeria spp. infection in broiler chickens** *Poult Sci* **95**:439–46 <https://doi.org/10.3382/ps/pev352> | PubMed | Google Scholar
57. Arendt MK, Knoll LJ, Cook ME (2019) **Oral antibody to interleukin-10 receptor 2, but not interleukin-10 receptor 1, as an effective Eimeria species immunotherapy in broiler chickens** *Poult Sci* **98**:3471–80 <https://doi.org/10.3382/ps/pez064> | PubMed | Google Scholar

58. Lessard PA, Parker M, Bougri O, Shen B, Samoylov V, Broomhead J, et al. (2020) **Improved performance of Eimeria-infected chickens fed corn expressing a single-domain antibody against interleukin-10** *Nat Food* **1**:119–26 <https://doi.org/10.1038/s43016-020-0029-6> | PubMed | Google Scholar
59. Balic A, Garcia-Morales C, Vervelde L, Gilhooley H, Sherman A, Garceau V, et al. (2014) **Visualisation of chicken macrophages using transgenic reporter genes: insights into the development of the avian macrophage lineage** *Development* **141**:3255–65 <https://doi.org/10.1242/dev.105593> | PubMed | Google Scholar
60. Whyte J, Glover JD, Woodcock M, Brzezczynska J, Taylor L, Sherman A, et al. (2015) **FGF, Insulin, and SMAD Signaling Cooperate for Avian Primordial Germ Cell Self-Renewal** *Stem Cell Reports* **5**:1171–82 <https://doi.org/10.1016/j.stemcr.2015.10.008> | PubMed | Google Scholar
61. Clinton M (1994) **A rapid protocol for sexing chick embryos (Gallus g. domesticus)** *Anim Genet* **25**:361–2 <https://doi.org/10.1111/j.1365-2052.1994.tb00374.x> | PubMed | Google Scholar
62. Labun K, Montague TG, Krause M, Torres Cleuren YN, Tjeldnes H, Valen E (2019) **CHOPCHOP v3: expanding the CRISPR web toolbox beyond genome editing** *Nucleic Acids Res* **47**:W171–W4 <https://doi.org/10.1093/nar/gkz365> | PubMed | Google Scholar
63. Concordet JP, Haeussler M (2018) **CRISPOR: intuitive guide selection for CRISPR/Cas9 genome editing experiments and screens** *Nucleic Acids Res* **46**:W242–W5 <https://doi.org/10.1093/nar/gky354> | PubMed | Google Scholar
64. Idoko-Akoh A, Taylor L, Sang HM, McGrew MJ (2018) **High fidelity CRISPR/Cas9 increases precise monoallelic and biallelic editing events in primordial germ cells** *Sci Rep* **8**:15126 <https://doi.org/10.1038/s41598-018-33244-x> | PubMed | Google Scholar
65. Ran FA, Hsu PD, Wright J, Agarwala V, Scott DA, Zhang F (2013) **Genome engineering using the CRISPR-Cas9 system** *Nat Protoc* **8**:2281–308 <https://doi.org/10.1038/nprot.2013.143> | PubMed | Google Scholar
66. Aunin E, Bohme U, Blake D, Dove A, Smith M, Corton C, et al. (2021) **The complete genome sequence of Eimeria tenella (Tyzzer 1929), a common gut parasite of chickens** *Wellcome Open Res* **6** <https://doi.org/10.12688/wellcomeopenres.17100.1> | PubMed | Google Scholar
67. Johnson J, Reid WM (1970) **Anticoccidial drugs: lesion scoring techniques in battery and floor-pen experiments with chickens** *Exp Parasitol* **28**:30–6 [https://doi.org/10.1016/0014-4894\(70\)90063-9](https://doi.org/10.1016/0014-4894(70)90063-9) | PubMed | Google Scholar
68. Nolan MJ, Tomley FM, Kaiser P, Blake DP (2015) **Quantitative real-time PCR (qPCR) for Eimeria tenella replication--Implications for experimental refinement and animal welfare** *Parasitol Int* **64**:464–70 <https://doi.org/10.1016/j.parint.2015.06.010> | PubMed | Google Scholar

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Reviewer #1 (Public review):

Summary:

In this study, Meunier et al. investigated the functional role of IL-10 in avian mucosal immunity. While the anti-inflammatory role of IL-10 is well established in mammals, and several confirmatory Knock-out models available in mice, IL-10's role in avian mucosal immunity is so far correlative. In this study the authors generated two different models of IL-10 ablation in Chickens. A whole body knock-out model, and an enhancer KO model leading to reduced IL10 expression. The authors first performed in vitro LPS stimulation based experiments, and then in vivo two different infection models employing *C. jejuni* and *E. tenella*, to demonstrate that complete ablation of IL10 leads to enhanced inflammation related pathology and gene expression, and enhanced pathogen clearance. At a steady-state level, however, IL-10 ablation did not lead to spontaneous colitis.

Strengths:

Overall the study is well executed and establishes an anti-inflammatory role of IL-10 in birds. While the results are expected, and not surprising, this appears to be the first report to conclusively demonstrate IL-10's anti-inflammatory role upon its genetic ablation in avian model. Provided the applicability of this information in combating pathogen infection in livestock species in sustainable industries like poultry, the study is worth publishing.

Weaknesses:

The study is primarily a confirmation of the already established anti-inflammatory role of IL-10.

Comments on revised version:

The authors have incorporated most of the points raised, and provided a reasonable argument for not considering DSS mediated colitis as an additional model.

<https://doi.org/10.7554/eLife.106252.2.sa2>

Reviewer #2 (Public review):

Summary:

The authors were to investigate functional role of IL10 on mucosal immunity in chickens. CRISPR technology was employed to generate IL10 knock out chickens in both exon and

putative enhancer regions. IL10 expressions were either abolished (knockout in exon) or reduced (enhancer knock-out). IL-10 play an important role in the composition of the caecal microbiome. Through various enteric pathogens challenge, deficient IL10 expression was associated with enhanced pathogen clearance, but with more severe lesion score and body weight loss.

Strengths:

Both in vitro and in vivo knock-out in abolished and reduced IL10 expression and broad enteric pathogens were challenged in vivo and various parameters were examined to evaluate the functional role of IL10 on mucosal immunity.

Weaknesses:

Overexpression of IL10 either in vitro or in vivo may further support the findings from this study.

Comments on revised version:

The authors' response and justifications are appropriate.

<https://doi.org/10.7554/eLife.106252.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

*In this study, Meunier et al. investigated the functional role of IL-10 in avian mucosal immunity. While the anti-inflammatory role of IL-10 is well established in mammals, and several confirmatory knockout models are available in mice, IL-10's role in avian mucosal immunity is so far correlative. In this study, the authors generated two different models of IL-10 ablation in Chickens. A whole body knock-out model and an enhancer KO model leading to reduced IL10 expression. The authors first performed in vitro LPS stimulation-based experiments, and then in vivo two different infection models employing *C. jejuni* and *E. tenella*, to demonstrate that complete ablation of IL10 leads to enhanced inflammation-related pathology and gene expression, and enhanced pathogen clearance. At a steady-state level, however, IL-10 ablation did not lead to spontaneous colitis.*

Strengths:

Overall, the study is well executed and establishes an anti-inflammatory role of IL-10 in birds. While the results are expected and not surprising, this appears to be the first report to conclusively demonstrate IL-10's anti-inflammatory role upon its genetic ablation in the avian model. Provided this information is applicable in combating pathogen infection in livestock species in sustainable industries like poultry, the study will be of interest to the field.

Weaknesses:

The study is primarily a confirmation of the already established anti-inflammatory role of IL-10.

We do not agree that this work is primarily confirmatory. The anti-inflammatory role of IL10 was indeed known previously from studies in mammals. The much more general insight from the current study is our demonstration of the intrinsic trade-off between inflammation and tolerance in the response to both the microbiome (which was significantly altered in the IL10 knockout birds) and mucosal pathogens. The study of *Eimeria* challenge in particular highlights the fact that it may be better for the host to tolerate a potential pathogen than to take on the cost of elimination.

Reviewer #2 (Public review):

Summary:

The authors were to investigate the functional role of IL10 on mucosal immunity in chickens. CRISPR technology was employed to generate IL10 knock-out chickens in both exon and putative enhancer regions. IL10 expressions were either abolished (knockout in exon) or reduced (enhancer knock-out). IL-10 plays an important role in the composition of the caecal microbiome. Through various enteric pathogen challenges, deficient IL10 expression was associated with enhanced pathogen clearance, but with more severe lesion scores and body weight loss.

Strengths:

Both in vitro and in vivo knock-out abolished and reduced IL10 expression, and broad enteric pathogens were challenged in vivo, and various parameters were examined to evaluate the functional role of IL10 on mucosal immunity.

Weaknesses:

Overexpression of IL-10 either in vitro or in vivo may further support the findings from this study.

An overexpression experiment, regardless of outcome, would not necessarily support or invalidate the findings of the current study. It would address the question of whether the absolute concentration of IL10 produced alters the outcome of an infection.

Reviewer #1 (Recommendations for the authors):

The following are the recommendations that, in my opinion, will be helpful to enhance the quality of the study.

Major point:

The authors at a steady state did not observe any sign of spontaneous colitis. Since IL-10 KO in mice leads to enhanced pathological score upon DSS-mediated induction of colitis, and several colitis models are well established in birds, it will be worthwhile to test the consequence of experimentally inducing colitis in this context.

One of the novel features of this study is the observation that the microbiome is modified in the IL10KO HOM chicks, which may serve to mitigate potential spontaneous pathology; we now mention this in the discussion. We agree that it could be worthwhile in the future to look at additional challenge models. However, we would argue that the *Eimeria* challenge is a sufficiently adequate experimentally-induced model of colitis to demonstrate the increased inflammation that occurs in an IL10-deficient bird. This is further supported by evidence of enhanced inflammatory responses in the caeca of IL10KO HOM birds challenged with *Campylobacter* or *Salmonella* relative to WT controls. See in the revised manuscript (pages 12-13).

Minor points:

(1) In Figure 2B, the authors should confirm whether the ROS-AV163 groups also have LPS treatment.

The legend for Figure 2B already states that neutralizing anti-IL10 antibody was added to LPS-stimulated BMDMs: “Nitric oxide production was assessed by measuring nitrite levels using Griess assay for LPS-stimulated BMDMs [...] in the absence or presence of neutralizing anti-IL10 antibody ROS-AV163”. However, for added clarity we have now modified the x-axis label for Figure 2B (“+ROS-AV163” replaced by “+LPS +anti-IL10”) and we have also made minor changes to the figure legend. See in the revised manuscript (page 33).

(2) In Figure 3F, the authors should discuss why the duodenum of KO birds has enhanced infiltration compared to WT?

We are not sure what the reviewer is referring to here. Although not specifically mentioned in Figure 3F, there is no statistically significant difference in cellular infiltration in the duodenum of IL10KO WT and HOM birds raised in our specified pathogen-free (SPF) facility, nor in the duodenum of IL10KO WT and HOM birds raised in our conventional facility (Mann-Whitney U tests, $p > 0.1$ in both cases); this can be seen in the sums of histopathological scores shown in Figures 3C (SPF facility) and 3E (conventional facility). Figure 3F shows that there is a statistically significant difference in cellular infiltration scores in the duodenum and proximal colon of both IL10KO WT and HOM birds based on the environment they are raised in (SPF vs conventional). We have made minor changes to the text to clarify this. See in the revised manuscript (page 7).

(3) The authors should discuss the observed differences in the *C. jejuni* colonization results among the two cohorts at week 1 and week 2 post-infection.

Numbers of *C. jejuni* in the caeca of IL10KO HOM birds were markedly lower than for WT controls at 1-week post-infection in cohort 1, and at both time intervals post-infection in cohort 2 (Figure 4A). This reached statistical significance at 1-week post-infection in cohort 1 and at 2-weeks post-infection in cohort 2. It is evident from Figure 4A that considerable inter-animal variance existed in each group, and in the IL10KO HOM birds in particular. This is typical of *C. jejuni* colonisation in chickens, where bacterial population structures have been reported to be variable and unpredictable (Coward et al., Appl Environ Microbiol 2008, PMID: 18424530). Similar variation between time intervals, birds and repeated experiments has been reported when evaluating vaccines against *C. jejuni* colonisation (e.g. Buckley et al., Vaccine 2010, PMID: 19853682; Nothaft et al., Front Microbiol 2021, PMID: 34867850). We performed two independent studies for this reason. Taken together, we consider that our data provide convincing evidence of elevated pro-inflammatory responses upon *C. jejuni* infection in IL10KO HOM birds relative to WT controls that associates with reduced bacterial burden. Our data is also consistent with a published observation that a commercial broiler line with low IL10 expression had correspondingly elevated expression of CXCL1-1, CXCL1-2 and IL-1b (Humphrey et al., mBio 2014, reference 33 in our original submission). We have added text to the discussion to capture the points above. See in the revised manuscript (page 13).

Reviewer #2 (Recommendations for the authors):

For the animal challenging experiments, both IL10KO HOM and IL10EnKO HOM chickens were used for Eimeria challenge, but not for Salmonella and Campylobacter. Could the authors justify why?

The *Eimeria* challenge produced a much higher and more reproducible level of inflammation than either of the bacterial challenge models. Within the parasite challenge cohorts, IL10KO HET and IL10EnKO HOM birds were only marginally different from WT controls (e.g. parasite replication: Figures 5A and B; lesion scores: Figures 5E and F; body weight gain: Figures 5G and H). Given the more limited response and the inter-individual variation in the bacterial challenge models, we felt that analysis of a sufficiently large cohort of the IL10KO HOM was appropriate, while additional cohorts of IL10KO HET and IL10EnKO HOM birds large enough to detect statistically significant differences could not be justified.

In the M&M, there was no mention of # of birds generated for IL10EnKO HOM, HET, etc.

Full details of bird numbers can be found in SI Appendix Table S1 “Number of IL10KO and IL10EnKO WT, HET and HOM chicks hatched in the NARF SPF chicken facility in the first (G1) and second (G2) generations”. Table S1 is already referred to in the Results section “Generation of IL10-deficient chickens”; we have now also clearly referred to it in the “Animals” and “Generation of surrogate host chickens and establishment of the IL10KO and IL10EnKO lines under SPF conditions” sections of the Materials and Methods. In all three sections we have also added some text to clarify that the table details G1 and G2 bird numbers. See in the revised manuscript (pages 5, 15, 17).

From the results of Campylobacter challenge, the results from the cohort 1 and cohort 2 were not consistent at both 1 and 2 weeks of post-infection. There is not much discussion on this inconsistency. What is the final conclusion: significant difference in week 1 or week 2, OR none of them, OR both of them. What would happen if an additional cohort were conducted for Salmonella and Eimeria?

As noted in response to Reviewer 1 (minor point 3), we have now added text to the discussion on the partial inconsistency between independent *C. jejuni* challenge studies. We do not feel that additional experiments to address this comment are required. Highly significant increases in the infiltration of lymphoplasmacytic cells and heterophils were detected in IL10KO HOM chickens relative to WT controls in the caeca, a key site of *Campylobacter* colonisation. This was consistently observed in two independent cohorts at both 1- and 2-weeks post-infection (SI Appendix Figures S7 and S8) and was reflected in similar patterns of expression of pro-inflammatory genes at these intervals in both cohorts (Figure 4B). As our laboratory has observed substantially less variation between repeated *Salmonella* challenges, a single study was performed, but with adequate power to detect statistical differences. The effects of *E. tenella* infection in IL10KO WT and HOM birds were replicated (compare Figure 4 with data from day 6 in Figure 5).

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